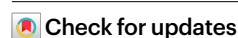


A dynamic sequence of visual processing initiated by gaze shifts

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Animals move their head and eyes as they explore the visual scene. Neural correlates of these movements have been found in rodent primary visual cortex (V1), but their sources and computational roles are unclear. We addressed this by combining head and eye movement measurements with neural recordings in freely moving mice. V1 neurons responded primarily to gaze shifts, where head movements are accompanied by saccadic eye movements, rather than to head movements where compensatory eye movements stabilize gaze. A variety of activity patterns followed gaze shifts and together these formed a temporal sequence that was absent in darkness. Gaze-shift responses resembled those evoked by sequentially flashed stimuli, suggesting a large component corresponds to onset of new visual input. Notably, neurons responded in a sequence that matches their spatial frequency bias, consistent with coarse-to-fine processing. Recordings in freely gazing marmosets revealed a similar sequence following saccades, also aligned to spatial frequency preference. Our results demonstrate that active vision in both mice and marmosets consists of a dynamic temporal sequence of neural activity associated with visual sampling.

Visual perception is an active process in which animals move through their environment and sample the visual scene through movements of the body, head and eyes. Self-motion and sampling behavior have a profound impact on the dynamic pattern of retinal visual input, which can play an integral role in processing and interpreting the visual scene^{1–4}. Correspondingly, a variety of movement-related signals are observed throughout the brain⁵. In the visual cortex, eye movement signals have been studied in nonhuman primates and cats^{6–10}, and recently in mice¹¹. Cortex-wide signals of locomotion and body/facial movements are present in head-fixed mice^{12–14}, and head rotation signals are seen in rodent visual cortex^{15–18}. Importantly, these signals can be attributed

to the movement itself, the consequence of movement on the visual input or a combination thereof.

However, the computational role of head/eye movement signals in natural vision remains enigmatic, particularly since they are often studied under constrained conditions of head or gaze fixation, with sparse, artificial visual stimuli. Likewise, it is not clear whether there are common principles of active visual processing across rodents and primates, given differences in eye movement repertoires and other visual specializations. Indeed, it has recently been shown that the gain modulation of mouse V1 neurons by locomotion¹² is present in primates, but differs in magnitude and sign¹⁹. We therefore sought to

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characterize the neural representation of visual sampling movements in mice and marmosets, under natural conditions of free movement and viewing, respectively.

By measuring both head and eye movements together with neural activity in freely moving mice, we found that most V1 neurons do not respond to compensatory head/eye movements but respond to gaze-shifting movements with a variety of response dynamics, resulting in a temporal sequence of population activity following a gaze shift. Under head-fixation, when presented with sequentially flashed full-field stimuli, responses in many neurons closely matched the response to gaze shifts, suggesting that the pattern of gaze-shift transients is largely, though not entirely, explained by the response to rapid onset of new visual input. The time course of responses corresponded predictably with preference for increasing spatial frequency (SF), consistent with the coarse-to-fine theory of visual processing²⁰. Notably, we observed a similar sequence of processing following saccades in head-fixed marmosets that were allowed to freely gaze while viewing natural scenes. These results provide insight into neural signals previously observed around head and eye movements and reveal a computational principle for coarse-to-fine visual processing during active vision in complex natural scenes that is shared across rodents and primates.

Results

V1 neurons respond specifically to gaze-shifting movements

First, we sought to characterize the responses of individual neurons in mouse V1 to large-amplitude head and eye movements. To monitor neural responses during free movement, we used a recording system (Fig. 1a) consisting of a head-mounted camera to record the position of the right eye, a forward-facing wide-angle (120°) camera to capture the visual scene, an inertial measurement unit (IMU) to record head orientation and acceleration, and a chronically implanted 64- or 128-channel linear silicon probe in V1 for single-unit electrophysiology (Fig. 1b)^{21,22}. To assess the impact of the recording system on behavior, we calculated the mean head pitch and roll, which revealed minimal effects on posture (Extended Data Fig. 1a,b).

Mice coordinate their head and eye movements in a ‘saccade-and-fixate’ pattern of selecting and stabilizing the input to the retina that is observed across the animal kingdom²³. When the mouse moves its head, the accompanying eye movements can be generally categorized into two functional types: compensatory or gaze-shifting^{21,24}. Gaze refers to the direction the eyes are pointing relative to the visual scene—the sum of head direction and eye position within the head—which determines the retinal input. Gaze-shifting eye movements accompany head movements in the same direction to change the focus of the animal’s gaze, and are typically abrupt, saccadic movements (Fig. 1c, left). Conversely, compensatory eye movements counteract head movements of the opposite direction to keep gaze stabilized during self-motion (through vestibulo-ocular and other reflexes) and are typically smooth in trajectory (Fig. 1c, right). Together, these two paired eye/head movements lead to a pattern of abrupt shifts of gaze (saccade) interspersed by periods of stabilized gaze (fixate) during continuous head movement. The vast majority of gaze-shifting movements in freely moving rodents are horizontal; vertical eye movements are generally compensatory^{11,21,24,25}.

As observed previously, large-amplitude horizontal head movements in our recordings were usually accompanied by a horizontal eye movement^{21,24}. The coordinated impact of gaze-shifting and compensatory movements is observed through gaze direction (sum of eye and head), showing a saccade-and-fixate pattern with periods of gaze stabilization (fixations) interspersed by abrupt shifts (saccades; Fig. 1d). This head movement categorization can be visualized in a plot of eye versus head velocity (Fig. 1e), where compensatory movements fall along the negative diagonal, since head and eyes move nearly equal but oppositely, while gaze shifts form a distinct off-diagonal pattern.

Based on this, we defined gaze-shifting movements as timepoints with gaze velocity $>240 \text{ deg s}^{-1}$, and compensatory as timepoints with gaze velocity $<120 \text{ deg s}^{-1}$, where gaze velocity is defined as the sum of the eye and head velocities (Methods). In both cases, we required a head velocity $>60 \text{ deg s}^{-1}$ to be considered a movement timepoint. During locomotion faster than 2 cm s^{-1} , the frequency of gaze-shifting movements doubled relative to slow and stationary periods (locomotion: $107 \pm 6 \text{ per min}$; slow and stationary periods: $50 \pm 3 \text{ per min}$; Extended Data Fig. 1c, left). A similar increase was observed for compensatory movements (locomotion: $361 \pm 11 \text{ per min}$; slow and stationary periods: $229 \pm 15 \text{ per min}$; Extended Data Fig. 1c, right). Thus, the difference in frequency of compensatory and gaze-shifting is not attributable to locomotion. Aligning gaze position to the time of gaze-shifting or compensatory movements reveals the characteristic step with gaze shifts and little change in gaze position for compensatory movements (Fig. 1f).

We extracted single-unit neural activity in freely moving mice around the time of each gaze-shifting or compensatory eye/head movement (Fig. 1g). This revealed striking responses following gaze-shifting movements, including positive, biphasic and negative responses. Notably, the same modulation is not generally found around compensatory movements. Across a population of 716 neurons in 9 mice, responses to gaze shifts were abundant (74.2%; 531 of 716) but compensatory responses were not (6.7%; 48 of 716; Fig. 1h). Threshold values to separate compensatory and gaze-shifting movements were chosen to isolate these two movement types based on previous findings²¹ (Extended Data Fig. 1d); however, we also explored intermediate gaze shifts (eye/head movements that shift the gaze $>120 \text{ deg s}^{-1}$ and $<240 \text{ deg s}^{-1}$; Extended Data Fig. 1e–g). Neuronal responses were largest to gaze shifts, with relatively low-amplitude responses to these intermediate movements (Extended Data Fig. 1h,i). Thus, V1 neurons predominantly respond to gaze-shifting eye/head movements, which produce abrupt changes in the retinal input, and do not generally respond to compensatory movements, which stabilize the retinal input. These findings support the notion that some component of previously observed responses in V1 to head movements may be attributed to an accompanying saccadic eye movement and the resulting change in visual input, rather than the head movement itself¹⁸.

Diversity and temporal sequence of gaze-shift responses

Neural responses to gaze shifts were diverse, both in sign of modulation and temporal dynamics. To characterize this diversity, we performed *k*-means clustering of units based on principal component analysis (PCA) of the normalized peri-event time histograms (PETHs) for gaze shifts (Extended Data Fig. 2a), with the onset of the event defined as the onset of the gaze shift, which will introduce a new visual stimulus onto the retina. For most units, the response for one direction of movement was stronger than the opposite direction. We determined their preferred direction as the direction of movement yielding the largest peak response and used this PETH for clustering. This clustering provides a means to describe the diversity within the population, but it is important to note that this is subdividing a continuous population. We selected a value of $k = 5$ clusters as it captured the diversity of responses while still providing distinct gaze-shift motifs in each cluster.

This resulted in four responsive clusters (Fig. 2a,b), which showed early positive (11.5% of population; Extended Data Fig. 2b), late positive (18.9%), biphasic (23.7%) and negative (9.2%) responses to gaze shifts, as well as a largely unresponsive fifth cluster (36.7%). Mean firing rates and the relative fraction of putative excitatory/inhibitory cells based on spike waveform were similar for each cluster (Extended Data Fig. 2d–f). Some units showed direction selectivity for leftward or rightward gaze shifts, especially in the early cluster (Extended Data Fig. 2g). Intriguingly, while most clusters were uniformly distributed in cortical depth, neurons in the early positive population were more likely to be located in the deep layers (Extended Data Fig. 2h).

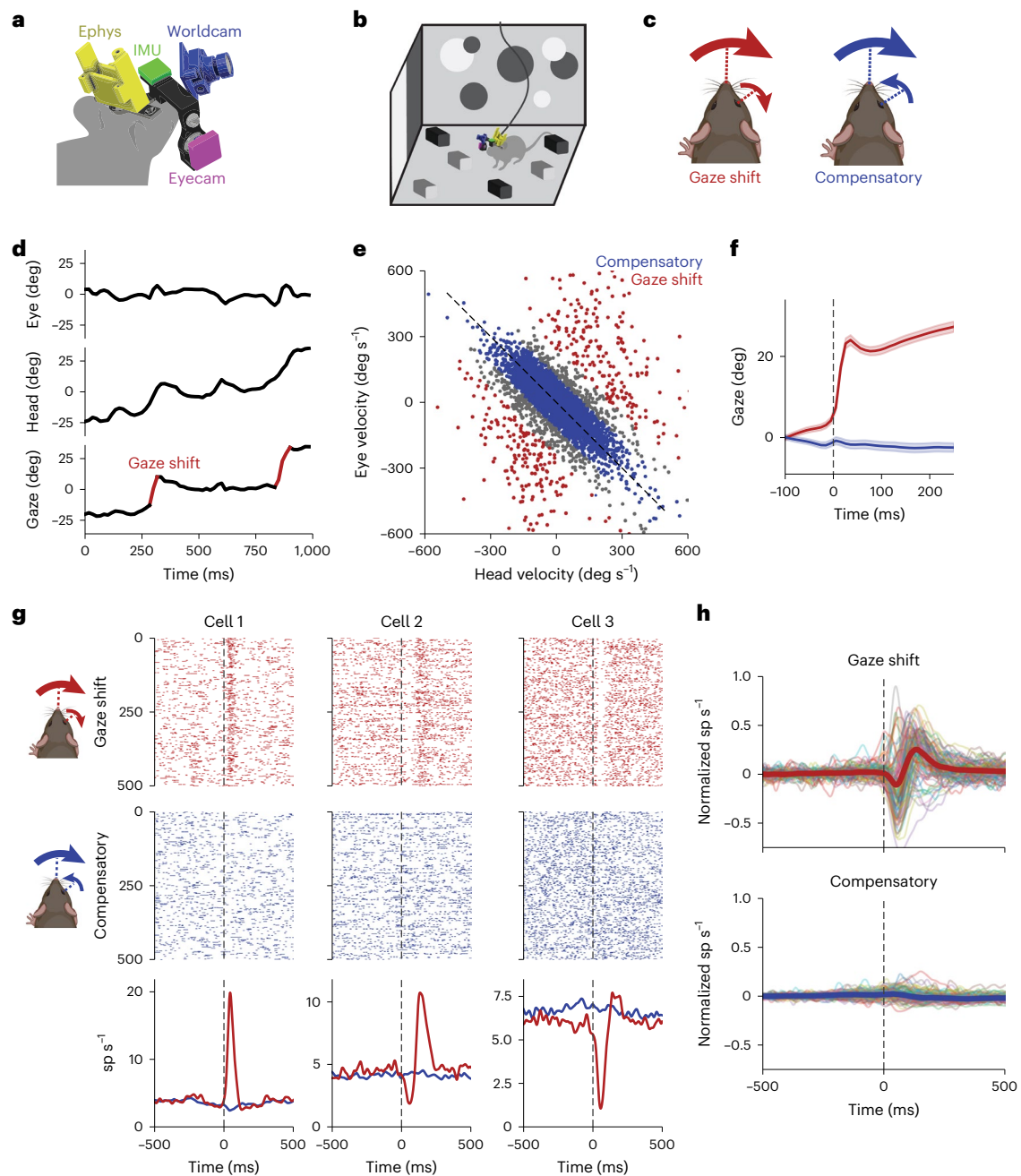


Fig. 1 | V1 neurons preferentially respond to gaze-shifting eye/head movements. **a**, Schematic of the head-mounted recording system. **b**, Experimental paradigm, in which a mouse freely moves through a visually complex environment. **c**, Schematic of gaze-shifting and compensatory eye/head movements, in which the eyes move either with or against the movement of the head. Note that the arrows are not to scale, and that eye versus head movement amplitudes are not always equal in either condition. **d**, Mice change their horizontal eye position (top) to compensate for changing horizontal head position (middle), punctuated by rapid gaze-shifting eye movements. The resulting gaze position (bottom), the sum of horizontal eye and head positions,

shows a 'saccade-and-fixate' pattern. **e**, Scatter plot of eye and head velocities subsampled (25×) from an example 64-min recording, showing segregation of compensatory versus gaze-shifting eye movements. **f**, Mean ± s.e.m. position of the animal's gaze around gaze-shifting (red) and compensatory (blue) movements in the leftward head direction across all recordings. **g**, Response of three example cells to 500 randomly sampled gaze-shifting (top) or compensatory (middle) eye/head movements as a spike raster. Bottom, gaze-shifting and compensatory PETHs. **h**, Normalized PETHs of gaze-shifting (top) and compensatory (bottom) eye/head movements for 100 example units with a baseline firing rate >2 Hz, with median of all cells ($n = 716$) overlaid.

These clusters also corresponded in a predictable manner to tuning for angular velocity of horizontal head rotation (Extended Data Fig. 2i). Early positive neurons were positively tuned for angular velocity, and direction-selective, while biphasic and negative clusters showed negative tuning, consistent with their response immediately after a gaze shift. On average, none of the clusters had a consistent response to compensatory movements (Fig. 2c), though most neurons that responded

to compensatory movements also responded to gaze shifts. However, biphasic/negative clusters had a greater number of compensatory responsive units than early/late clusters (Extended Data Fig. 2j–l).

Notably, the mean traces for each cluster (Fig. 2b) and the projection in principal component space (Extended Data Fig. 2a) suggested a temporal sequence, from early positive through negative. Sorting units based on the latency of their peak positive response to

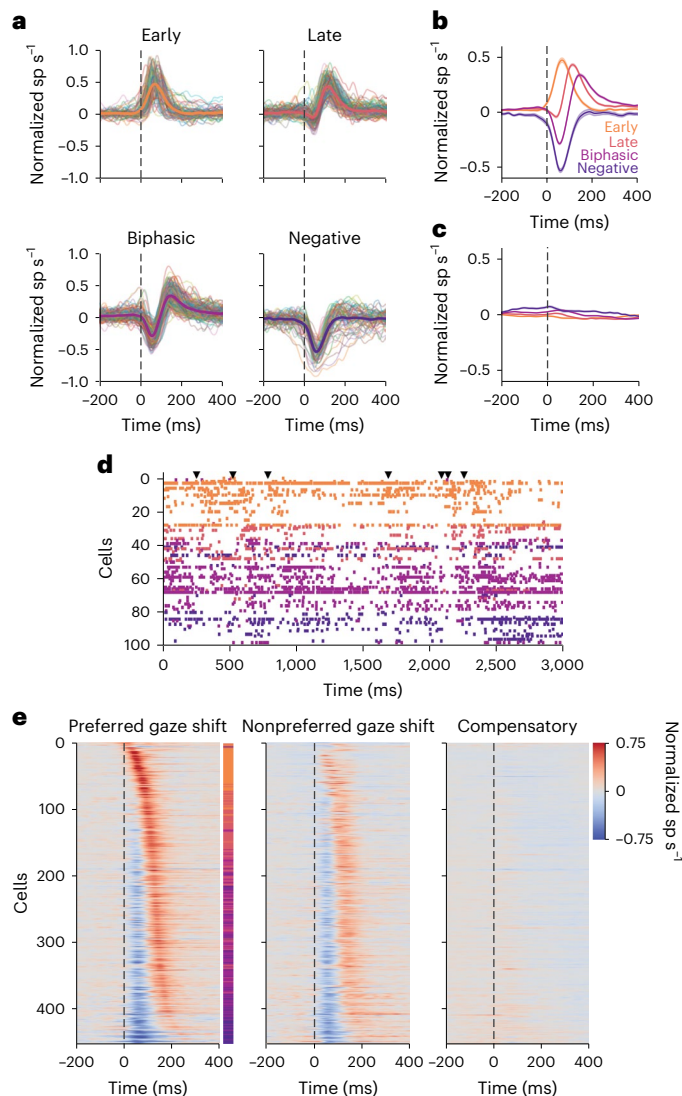


Fig. 2 | Diversity and temporal sequence of gaze-shift responses.

a, Normalized PETHs for preferred direction of gaze shifts in responsive units, grouped by *k*-means clustering, with each cluster's mean overlaid. *n* = 9 mice, 716 units (early = 82, late = 135, biphasic = 170, negative = 66, unresponsive = 263). **b**, Mean normalized PETH in response to gaze-shifting eye/head movements, clustered by response type. **c**, Same as **b** for compensatory movements with cells clustered on their gaze-shift response. **d**, Example spike rasters from a 3-s period of freely moving activity, showing all units responsive to gaze shifts (*n* = 98 of 128), color-coded by response type cluster. Gaze-shift onsets are indicated by black arrows above. **e**, Normalized PETHs for gaze-shifting movements in the preferred (left) and nonpreferred (middle) directions, and for compensatory movements (right). All responsive cells are shown, sorted along the *y* axis by response latency. The vertical colorbar to the side of the left panel shows each cell's assigned cluster using the colors in **a**.

gaze shifts revealed a continuous sequence of activation across the population following gaze shifts that was evident in single trial spike rasters (Fig. 2d and Supplementary Video 1). Applying this sorting to the mean PETHs (Fig. 2e) revealed a similar sequence, with positive peaks gradually becoming more delayed and an early negative component appearing (Fig. 2e, left). We confirmed that this sequence was not an artifact of sorting through cross-validation, which showed that both temporal order and peak response latencies (Extended Data Fig. 3a,b) were maintained when sorted on one half of the dataset (training set) and applied to the other half (test set; Pearson correlation coefficient, $r = 0.870$, $P = 2.51 \times 10^{-140}$). The temporal sequence pattern was present,

though weaker, in the nonpreferred direction (Fig. 2e, middle), and was absent for compensatory movements (Fig. 2e, right). Notably, the four responsive groups showed significantly different latencies to peak response (Extended Data Fig. 2c; $P < 0.05$ for all comparisons with no effect of experimental session; $P = 0.220$, linear mixed effects model) and generally maintained their grouping in the continuous sequence of temporal responses, with the early positive group showing the earliest peak positive response, followed by the late positive group, the biphasic group and the negative group (Fig. 2e, colorbar in left panel). Thus, the diverse responses to gaze shifts represent a temporal sequence across the population.

Temporal dynamics of gaze-shift responses require visual input

The selectivity of cells to gaze-shifting head and eye movements, but not compensatory movements, led us to hypothesize that visual input may predominantly drive the observed responses in many neurons, as opposed to primarily motor efference or vestibular signals. To test the role of visual input in generating gaze-shift responses, we recorded neural activity during free movement both in the light and in complete darkness.

We clustered cells using the gaze-shift PETHs measured under normal light conditions (Fig. 3a) and compared their responses in the light and dark (Fig. 3b). The distinct cluster responses present in the light were absent in the dark, with most clusters (late positive, biphasic and negative) showing an average suppression of activity with an onset before the start of the gaze shift (Fig. 3c; $n = 265$ units, 7 mice), consistent with previous reports of saccadic suppression²⁶. As in the light, neurons also showed little response to compensatory movements in the dark (Fig. 3d). Sorting units based on their response latency in the light revealed that the precise temporal sequence of responses found in the light was also not maintained in the dark (Fig. 3e). Together with the absence of modulation around compensatory movements, this suggests that the specific dynamic responses following a gaze shift rely on the corresponding change in visual input.

The mean response of the early positive group appeared flat in the dark, which resulted from a mix of units that showed suppression and a small fraction of units with strongly gaze direction-tuned responses that persisted in the dark (3.3% cells, $n = 9$ of 269; Extended Data Fig. 4). The latter may represent a small population that receives a direct corollary discharge signal for saccades, as found in previous work¹¹. However, for the vast majority of neurons, the diversity of responses we observed around gaze shifts in the light was replaced by a suppression in the dark that lacked a clear temporal sequence.

Flashed stimulus responses resemble gaze-shift responses

We sought to further test whether the observed responses to gaze shifts were visually driven by presenting stimuli that mimic the rapid onset of a new visual image that results from a gaze shift in a complex visual environment. Head-fixed mice were shown continuously flashed stimuli, including reversing checkerboard and sparse noise, before recording the same units during free exploration of the arena, and we compared the stimulus onset responses with gaze-shift responses.

Time-aligned rasters and PETHs for gaze shifts and reversing checkerboard stimulus are shown for example units in Fig. 4a. Notably, the visually evoked response to continuously flashed full-field stimuli resembled the neuron's response to gaze shifts, including positive, biphasic and negative responses. This similarity was evident at the population level as well, as the mean stimulus onset PETHs in response to the flashing checkerboard (Fig. 4b, left), clustered based on each unit's gaze-shift response type, closely matched responses to gaze shifts (Fig. 4c), as did mean responses to the shorter sparse noise stimulus (Fig. 4b, right; Fig. 4d). To quantify the similarity between gaze-shift responses and those to flashed stimuli, we computed the correlation coefficient between PETHs for gaze shifts and the flashed

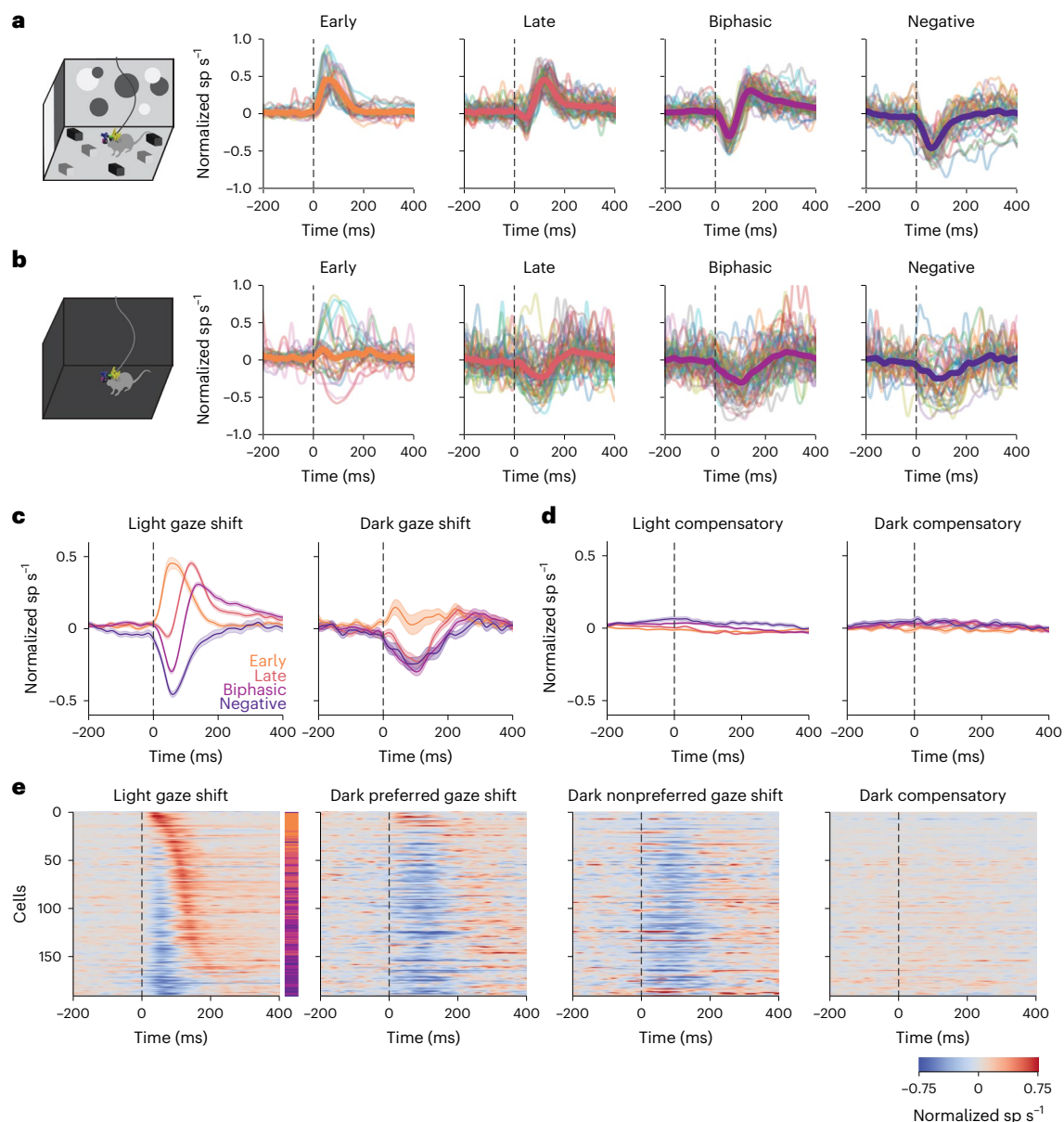


Fig. 3 | Temporal dynamics of gaze-shift responses depend on visual input. **a**, Normalized PETHs in light condition for the preferred direction of gaze shifts, clustered using *k*-means weights used for cells in Fig. 2. **n** = 7 mice, 269 units (early = 27, late = 56, biphasic = 66, negative = 42, unresponsive = 78). **b**, Same as **a** for dark condition. Cells are grouped by their cluster from **a**. **c**, Mean gaze-shift

responses of each cluster in light (left) and dark (right) conditions with standard error. **d**, Same as **c** for compensatory movements. **e**, Temporal sequence of responses to gaze shifts and compensatory movements in the light and dark. All temporal sequences are sorted by the cell's gaze-shift response latency for the light condition (left).

stimuli for each unit (0–250 ms following onset). This revealed that gaze-shifting and flashed stimulus PETHs were highly correlated for most units (Fig. 4e,f; checkerboard median = 0.64, *n* = 345; sparse noise median = 0.68, *n* = 263).

Furthermore, the temporal sequence of checkerboard and sparse noise responses, when sorted based on each unit's peak latency from freely moving gaze shifts, closely matched the temporal sequence from gaze shifts (Fig. 4g). Correspondingly, the latency of each unit's responses to head-fixed sequential stimuli was highly correlated with its latency for gaze-shift responses (Fig. 4h,i). These results cannot be explained by the mouse performing gaze shifts with stimulus onset, as mice made only infrequent eye movements during head-fixation (sparse noise: 11.0 ± 10.1 saccades per min; checkerboard: 13.6 ± 15.8 saccades per min). Interestingly, responses to head-fixed stimuli were slightly faster than responses to gaze shifts (checkerboard versus gaze

shift: slope = 0.55, intercept = 0.041; sparse noise versus gaze shift: slope = 0.44, intercept = 0.045). Although it is not apparent why this might be, one possibility is that the appearance of a flashed stimulus on the screen is abrupt, whereas saccades extended over tens of milliseconds, so the timepoint when the input is stabilized following a gaze shift may be substantially delayed relative to the gaze-shift onset. Regardless, the close correlation between responses to gaze shifts during free movement and responses to sequentially flashed stimuli strongly suggests that the temporal sequence of gaze-shift responses primarily results from large, rapid changes to visual input.

Coarse-to-fine processing around gaze shifts

We next sought to determine whether there was a relationship between the visual tuning properties of neurons and their temporal responses to gaze shifts. We presented drifting sinusoidal gratings at three spatial

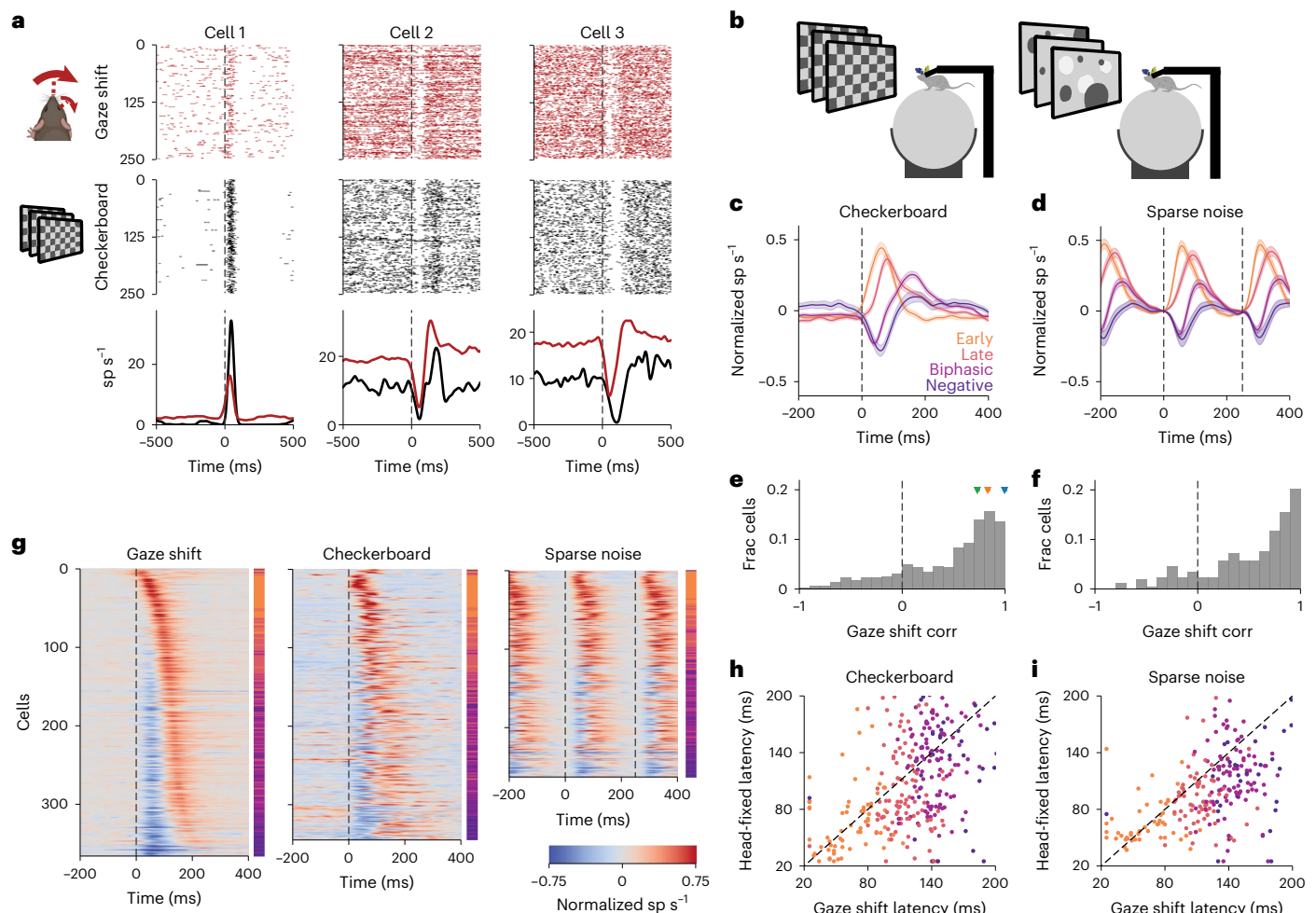


Fig. 4 | Head-fixed flashed stimulus responses resemble freely moving gaze-shift responses. **a**, Three example cells with positive (left), biphasic (middle) and negative (right) responses to freely moving gaze shifts. Spike rasters at the time of 250 randomly sampled gaze shifts during free movement (top) and 250 full-field reversals of a black and white checkerboard during a head-fixed recording of the same cell (middle). Bottom, gaze-shift (red) and checkerboard (black) PETHs for each cell. **b**, Schematic of head-fixed flashed checkerboard and sparse noise stimuli. **c**, Mean $\pm$ s.e.m. normalized response of cells to head-fixed flashed checkerboard, clustered on their responses to freely moving gaze shifts. $n = 9$ mice, 716 cells (checkerboard responsive = 472 of 716, early = 67, late = 100, biphasic = 123, negative = 55, gaze-shift unresponsive = 127). **d**, Same as **c** for sparse noise stimulus (sparse noise responsive = 333 of 716, early = 51, late = 69, biphasic = 99, negative = 44, gaze-shift unresponsive = 70). **e**, Histogram of Pearson correlation coefficient between gaze-shift PETH and flashed

checkerboard, plotted as fraction of cells (frac) versus gaze shift correlation (corr), (median = 0.64, $n = 345$; arrows indicate correlation, for example, cells 1 (blue, 0.99), 2 (orange, 0.83) and 3 (green, 0.73)) in **a**. **f**, Same as **e** for flashed sparse noise stimulus (median = 0.68, $n = 263$). **g**, Temporal sequence of gaze-shift responses (left, cells responsive to gaze shifts and either of the flashed head-fixed stimuli, $n = 366$), checkerboard (middle, cells responsive to both gaze shift and checkerboard, $n = 345$) and sparse noise (right, cells responsive to both gaze shift and sparse noise, $n = 263$) sorted on the latency of gaze-shift responses in freely moving conditions. **h**, Latency of positive peak in gaze-shifting responses versus head-fixed stimulus responses for checkerboard (Pearson correlation coefficient, $r = 0.487$, $P = 5.89 \times 10^{-22}$, slope = 0.55, intercept = 0.041). **i**, Same as **h** for sparse noise stimulus (Pearson correlation coefficient, $r = 0.526$, $P = 4.19 \times 10^{-20}$, slope = 0.44, intercept = 0.045).

(0.02, 0.08, 0.32 cycles per deg, or cpd) and two temporal (2, 8 cycles per s) frequencies to head-fixed mice before recording of the same units during free movement (Fig. 5a,b). When organized by gaze-shift response type, the mean PETHs for gaze-shift clusters to drifting gratings reveal a large-amplitude onset transient in the positive clusters, and a smaller, slower rise time onset transient with a larger sustained response in the biphasic and negative clusters (Extended Data Fig. 5a,b). Note that the temporal dynamics of these responses differ from those to gaze shifts and flashed stimuli due to the continuously moving drifting gratings.

To compare the relative SF preference across the response sequence, we performed four separate analyses. First, we calculated mean tuning curves for each gaze-shift cluster, which showed a clear progression shifting from lower to higher SFs across clusters (Fig. 5c).

The early positive group was biased toward lower and mid SFs, which gradually shifted to a peak at the highest SFs through the biphasic and negative groups (multivariate analysis of variance (ANOVA), cluster $\times$ SF tuning, $F = 9.20$, $P = 8.71 \times 10^{-10}$). Calculating the preferred SF for units meeting a $\geq 2:1$ selectivity index also revealed corresponding differences in SF preference across clusters (Extended Data Fig. 5c), though this progression was somewhat weaker as this is a categorical metric and excludes a substantial number of neurons. We also computed an SF preference for each neuron based on the weighted average of its response to the three drifting gratings SFs (Fig. 5d), which provides a metric to capture the central tendency of the individual neuron SF tuning curves. This metric also showed an increase from lower to higher SFs across the four groups. This relationship was not due to differences in sampling across recording sessions (linear mixed effects model, $P = 0.247$

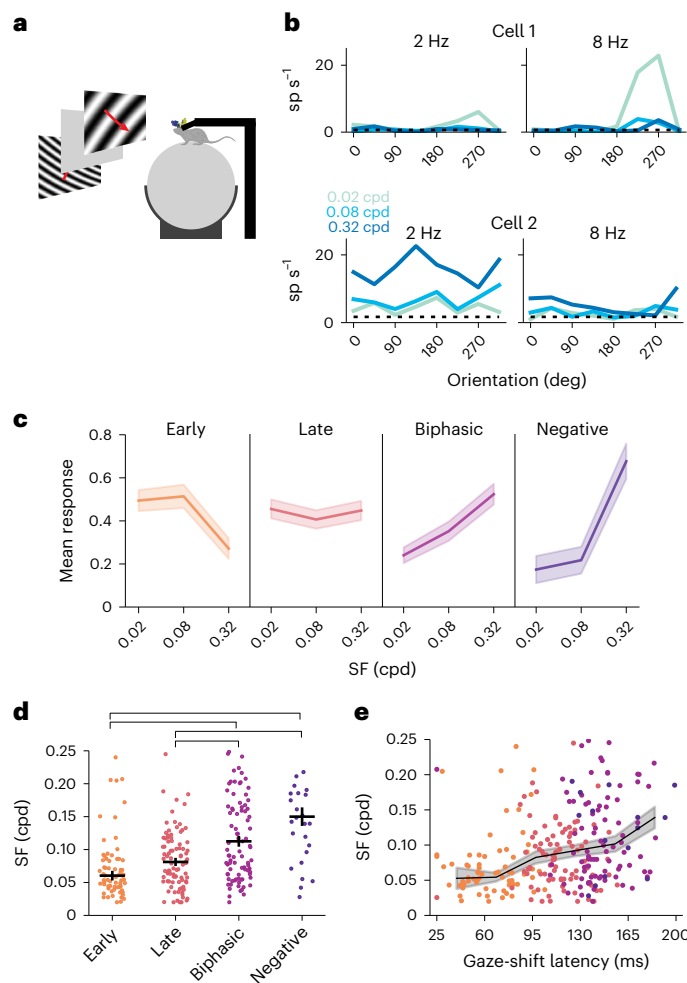


Fig. 5 | SF tuning demonstrates coarse-to-fine processing around gaze shifts. **a**, Schematic of head-fixed drifting gratings recordings. **b**, Orientation tuning curves for two example cells from head-fixed drifting gratings stimulus. Responses are shown for eight orientations of gratings (0 deg is the horizontal rightwards direction), three SFs, and low (left) and high (right) TFs. Spontaneous firing rates are shown as a horizontal dotted black line. **c**, Mean $\pm$ s.e.m. SF tuning curve for gratings-responsive cells in each cluster (multivariate two-way ANOVA, SF $\times$ cluster, $F = 9.20$, $P = 8.71 \times 10^{-10}$; $n = 9$ mice, grating responsive cells = 384 of 716, early = 71, late = 96, biphasic = 98, negative = 29, unresponsive = 90). **d**, SF preference for gratings-responsive cells in each gaze-shift response cluster, calculated as a weighted mean of responses. Median and standard error are shown for each cluster. Bars above indicate statistical significance at $P < 0.05$ (linear mixed effects model, $n = 9$ mice, $n = 384$ cells; early versus late $P = 0.145$, early versus biphasic $P = 3.04 \times 10^{-7}$, early versus negative $P = 5.49 \times 10^{-6}$, late versus biphasic $P = 4.54 \times 10^{-5}$, late versus negative $P = 3.96 \times 10^{-4}$, biphasic versus negative $P = 0.620$). Of 384 cells, 20 had a weighted mean SF response greater than 0.25 cps and are not visible in the scatter plot but were included in the calculation of median. **e**, Weighted SF preference versus gaze-shift response latency, for all cells responsive to gratings. Running median $\pm$ s.e.m. for all cells is overlaid. The color of each point indicates the cluster from gaze-shift responses (Pearson correlation coefficient, $r = 0.286$, $P = 1.73 \times 10^{-6}$).

grouped by subject). Finally, when sorting units based on gaze-shift response latency, the median SF preference increases as a function of latency (Fig. 5e). Together, these different analyses demonstrate that the temporal sequence of responses following gaze shifts corresponds to a progression from lower-SF-preferring neurons to higher-SF-preferring neurons. These results are consistent with the coarse-to-fine theory of visual processing, which posits that the spatial components of new visual input are processed sequentially from lower to higher SFs²⁰.

Distinct temporal frequency tuning for early/late responses

The relationship between gaze-shift response and weighted temporal frequency (TF) preference was qualitatively opposite to that of SF. The early positive group preferred the highest TF across the population, while the three later groups preferred relatively low TFs (Extended Data Fig. 5d,e; multivariate ANOVA, TF $\times$ cluster $F = 21.45$, $P = 3.45 \times 10^{-13}$). Correspondingly, the median TF preference decreased as a function of increasing response latency (Extended Data Fig. 5f). This relationship was not due to differences in sampling across recording sessions (linear mixed effects model, $P = 0.659$ grouped by subject). Calculating the preferred TF of units meeting a $\geq 2:1$ selectivity index, as an alternative to the weighted TF preference metric, revealed a shift in the bias of TF tuning from early to later responses (Extended Data Fig. 5g). The observed relationship between TF preference and response latencies provides a potential explanation for the sequence we observe following gaze shifts: cells that prefer higher TF values respond quickly to gaze shifts, corresponding to the initial and abrupt change in visual input as a result of a shifting gaze. Cells that prefer lower TF values, conversely, respond later after the gaze shifts during gaze fixation and maintenance, when the stimulus is not rapidly changing.

A simple linear–nonlinear model provided support for this intuition (Supplementary Text and Extended Data Fig. 6) by demonstrating that temporal kernels corresponding to low and high TF can qualitatively reproduce the different response profiles observed. This model also provides an explanation for how these response profiles differ from responses to standard visual stimuli that contain a gray interstimulus interval (ISI), since new stimuli arrive continuously without an opportunity for neurons to return to baseline.

Coarse-to-fine temporal sequence in freely gazing marmosets

Given the finding of a coarse-to-fine visual processing sequence following gaze shifts in freely moving mice, we wondered whether a similar temporal sequence occurs in V1 of nonhuman primates during active visual sampling. Mouse vision differs in several ways from nonhuman primates, but particularly striking is the lack of a fovea and high-acuity vision^{27,28}. We therefore recorded the activity of single units in foveal V1 of head-fixed marmosets to determine the neural response to saccades across complex visual scenes.

Marmosets were first head-fixed and allowed to freely view natural image stimuli (Fig. 6a). Aligning spike times to the onset of saccades revealed gaze-shift responses in marmosets that were similar to those in freely moving mice. Spike rasters and PETHs from example units show positive, biphasic and negative responses following saccades (Fig. 6b). We clustered responsive units based on their normalized gaze-shift PETHs (k -means, $k = 4$ as the cluster of nonresponsive units was removed based on set criteria; Extended Data Fig. 7a). This resulted in qualitatively similar groups to those identified in mice (Fig. 6c,d and Extended Data Fig. 7b,c). We also found a similar temporal sequence of responses across the marmoset V1 population after sorting by latency of peak positive response (Fig. 6e), with the clusters maintaining their groupings across the temporal sequence in the same order as the mouse V1 units. We confirmed the sequence and latencies (Extended Data Fig. 3c,d) by cross-validation.

To determine whether this temporal sequence represents a similar coarse-to-fine mechanism, we examined SF tuning based on rapidly flashed sinusoidal gratings, presented while the monkey performed saccades to forage for embedded Gabor patches²⁹ (Fig. 7a and Methods). This dataset was originally designed to study neural signals associated with visual foraging, but for the present study the same data provide an accurate estimate of SF preference for the same neurons as in the recordings during natural image viewing. Specifically, the sinusoidal gratings represent a Hartley subspace that allows rapid mapping of receptive fields²⁹ as well as orientation and SF tuning, though not TF tuning.

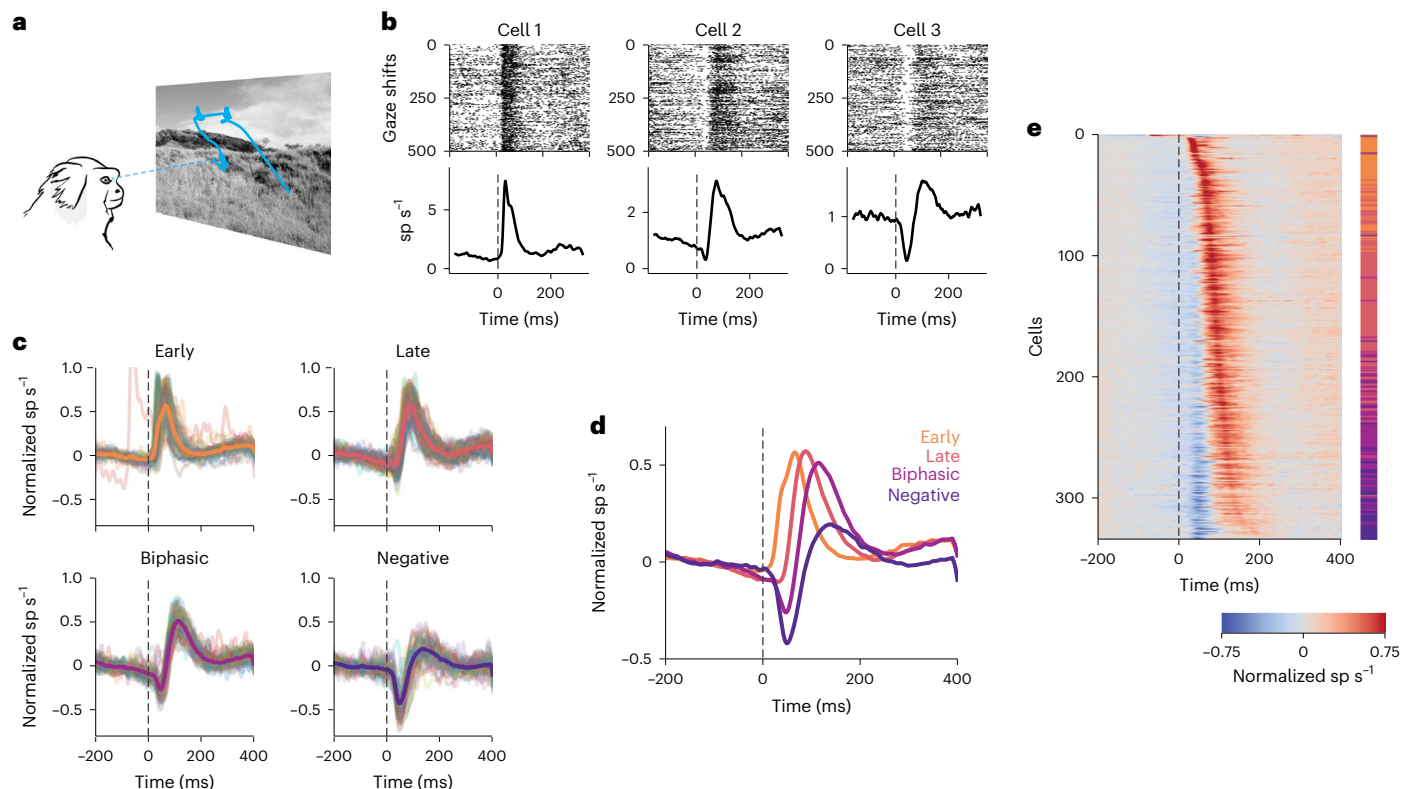


Fig. 6 | A similar temporal sequence of V1 saccade responses in freely gazing marmosets. a, Schematic of marmoset natural image free-viewing. **b**, Three example cells' spike rasters (top) and PETHs (bottom) for 500 randomly selected saccades. **c**, Normalized PETHs for saccades, grouped by *k*-means clustering, with each cluster's mean overlaid. *n* = 2 marmosets, 334 units (early = 64,

late = 136, biphasic = 84, negative = 50). **d**, Mean normalized saccade response of cells in each cluster. **e**, Normalized saccade PETHs sorted along the y-axis by positive peak latency. The vertical colorbar indicates the response cluster for each cell using the colormap from **c**.

Similar to the SF tuning of the gaze-shift response clusters in mice, marmoset neurons with earlier, positive saccade responses had tuning curves biased to lower SF preferences, centered around 2 cpd, and biphasic/negative were biased to higher SFs, centered around 4 cpd (Fig. 7b; multivariate ANOVA, $SF \times \text{cluster } F = 12.43$, $P = 1.82 \times 10^{-18}$). We observed a similar progression across the groups based on the weighted mean SF preferences (Fig. 7c). This relationship was not due to differences in sampling across recording sessions (linear mixed effects model, $P = 0.534$ grouped by subject). Likewise, we observed an increase in median weighted SF preference as a function of saccade response latency (Fig. 7d). Finally, calculating the peak SF for each unit showed a shift from 2 to 4 cpd across the four clusters (Extended Data Fig. 7d). We thus find that marmoset foveal V1 shows a temporal sequence of gaze-shift responses similar to that of freely moving mice, consistent with a coarse-to-fine pattern of visual processing shared across mouse and primate V1.

Discussion

We recorded the activity of V1 neurons as mice moved freely in an arena to determine how visual processing is influenced by eye and head movements at the cortical level. A wide scope of previous work has shown that activity of cortical neurons is correlated with movement, with V1 neurons showing responses to locomotion, head movements and eye movements. We found a diverse array of responses to gaze-shifting, but not compensatory, head and eye movements in V1 that together formed a temporal sequence of activity. When the animal moved in an arena lacking visual input (complete darkness), the continuous temporal sequence of responses was not maintained, and most neurons showed weak suppression of activity around the time of a gaze shift. Further, we found V1 neurons mimic their response to gaze shifts when shown

full-field sequentially flashed stimuli under head-fixation despite no corresponding head or eye movements. These findings suggest that a substantial component of the responses to gaze-shifting head and eye movements results from changes to visual input as opposed to motor efference or nonvisual sensory reafferent signals. We did observe evidence of nonvisual signals from responses in darkness—a small set of 'early positive' responses (3.3% of the population) and a common slow suppression signal. These signals could be combined, either additively or nonlinearly, with visual input in the light. A previous study reported a higher incidence of V1 neurons with apparent motor efference responses resulting from saccades (~35% direction-selective saccade responses¹¹). The discrepancy in the prevalence of these signals could be due to differences in analysis, such as threshold and normalization methods, or differences in stimulus conditions (recordings performed with the mouse in front of a gray screen versus in darkness here). A recent study reported a 9.5% incidence of saccade-responsive neurons in V1 (ref. 25).

Additionally, we found that neurons responding earlier to a gaze shift preferred higher TFs and lower SFs, and neurons responding later following a gaze shift preferred lower TFs and higher SFs. These findings are consistent with the coarse-to-fine model of visual processing, in which the visual system responds to coarse aspects of the visual scene before finer details. The temporal sequence that we observe is locked to the 'saccade-and-fixate' pattern that occurs during natural vision, during which rapid saccadic eye movements are interspersed with slow fixational movements in a pattern that formats the visual input arriving on the retina¹. Our findings therefore reflect the fact that natural visual input is neither continuous nor organized into temporally distinct trials as in highly controlled experimental paradigms. This impacts both the statistics of the visual input and the corresponding neural

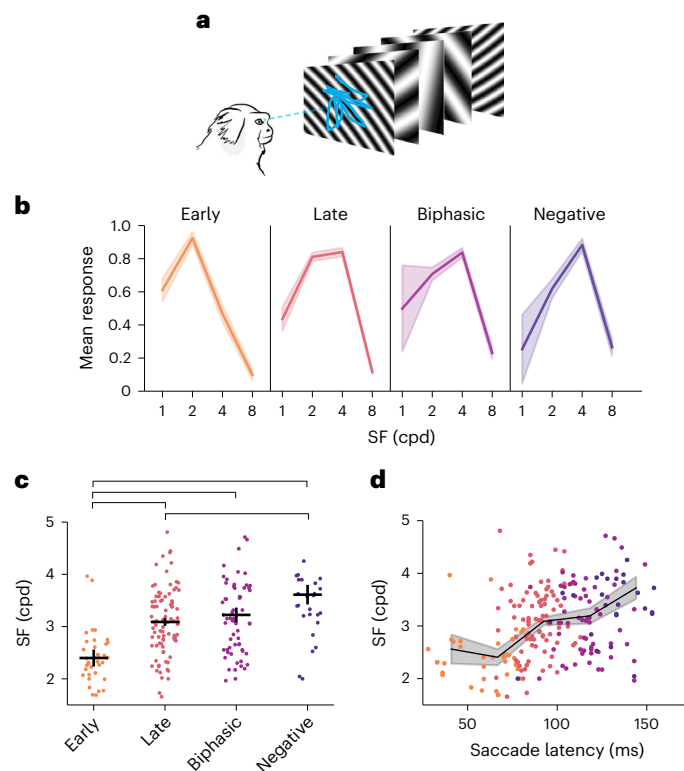


Fig. 7 | A coarse-to-fine temporal sequence in freely gazing marmosets.

a, Schematic of grating stimulus. Sinusoidal gratings of varying SF and orientation were presented rapidly (60 Hz) while monkeys performed saccades to forage for an embedded Gabor patch. **b**, Mean $\pm$ s.e.m. SF tuning curve for gratings-responsive cells in each cluster (multivariate two-way ANOVA, $SF \times \text{cluster } F = 12.43, P = 1.82 \times 10^{-18}, n = 2$ marmosets, $n = 238$ cells, early = 41, late = 88, biphasic = 70, negative = 39). **c**, Weighted SF preference for each saccade response cluster, calculated as a weighted mean of responses for four presented SFs. Median and standard error are shown for each cluster. Only gratings-responsive cells are shown. Bars above indicate statistical significance at $P < 0.05$ (linear mixed effects model, $n = 2$ marmosets, $n = 238$ cells; early versus late $P = 5.77 \times 10^{-3}$, early versus biphasic $P = 1.29 \times 10^{-4}$, early versus negative $P = 9.71 \times 10^{-6}$, late versus biphasic $P = 0.126$, late versus negative $P = 0.013$, biphasic versus negative $P = 0.253$). **d**, Weighted SF preference versus saccade response latency. Running median $\pm$ s.e.m. is overlaid. Point colors indicate response clusters. Outlier cells with a response latency below 25 ms ($n = 1$) or above 160 ms ($n = 12$) are not shown for plotting purposes but were included in statistical tests. $n = 334$ units (early = 40, late = 88, biphasic = 69, negative = 29; $r = 0.313, P = 8.54 \times 10^{-7}$).

dynamics, underscoring the need to study visual processing under naturalistic conditions.

These findings are remarkably consistent across mice and marmosets, both in terms of the specific temporal response profiles that we observe, and the correspondence with SF preference. This is notable since although many basic aspects of cortical visual processing are shared across mammals, one might expect that aspects related to active vision could differ. Primates have a pronounced foveal specialization accompanied by an expanded repertoire of eye movements, including voluntary targeted saccades independent of head movements. On the other hand, mice have less pronounced retinal specializations^{30,31}, and eye movements are largely coupled to head movement and image stabilization, with little evidence of independently targeted saccades^{21,24,32,33}. Nonetheless, when analyzed in a shared framework of gaze shifts versus fixations, a similar computational principle emerges—a temporal sequence of response dynamics corresponding to coarse-to-fine processing.

Temporal dynamics and gaze-shift responses

Our results provide insight into the nature of neural signals measured around head and eye movements in natural conditions. In one of the few studies of free-viewing of natural scenes in primates, a mix of positive and negative responses was observed following saccades³⁴. Likewise, neural responses around microsaccades, when the fovea moved within the stimulus, revealed positive, negative and biphasic responses⁷. Our results suggest that these diverse responses may correspond to the different temporal tuning of V1 neurons, which determines their response when the retinal input shifts abruptly following saccades or microsaccades. Likewise, a recent study observed modulation of V1 activity around head-orienting movements in rats¹⁸, with a diversity of positive and negative tuning in light and dark, with no systematic relationship. By measuring eye movements along with head movements, we found that most neurons primarily respond to the instants within an ongoing head movement that are associated with a saccadic movement of the eyes, which result in a gaze shift. Furthermore, our results explain the logic behind the diversity of head movement responses, based on the temporal sequence relative to gaze shifts and the SF/TF preferences measured with grating stimuli—from early positive neurons with lower SF/higher TF preference to biphasic and then negative neurons with higher SF/lower TF preference (Figs. 5 and 7 and Extended Data Fig. 2c). This time course also predicts responses in the dark, as all but the earliest positive neurons shift to suppression in the dark (Fig. 3).

It is striking that we see relatively little response to head movements that are not accompanied by a gaze shift, especially since a subset of V1 neurons show vestibular coding during head rotation in head-fixed subjects^{15,17}. However, in comparing head-fixed versus freely moving conditions, it is important to note that studies of coding in the vestibular nuclei have shown that the vestibular reafference resulting from active, volitional rotation (as in free movement) is canceled out, while in passive rotation (as in head-fixed conditions) it is not^{35,36}. Together, these data suggest that V1 may similarly differentiate between active and passive head movements, potentially by inheriting these signals from the vestibular nuclei.

Surprisingly, a large fraction of neurons had a negative response to gaze shifts in the light and to flashed stimuli. Negative responses in visual neurons around the time of a saccade have long been hypothesized to be suppression of visual perception while performing an eye movement across the visual scene. We find that such responses are common and represent part of a continuum from early positive responses through biphasic and finally negative responses. Notably, the neurons with the largest negative responses, and correspondingly latest positive responses, also preferred the lowest TFs. This suggests that the negative responses, at least in part, are a result of the neurons not responding to the high TF transient associated with the saccade, leading to a drop in activity, then slowly responding to the sustained visual input during fixation. This hypothesis is supported by a simple linear–nonlinear model (Supplementary Text), which recapitulates the diverse temporal dynamics, including negative responses, based purely on different temporal kernels for a neuron's response that correspond to higher and lower TF, or transient and sustained responses.

Notably, these temporal dynamics arise as a result of the abrupt arrival of new visual input, during both gaze shifts and continuously flashed stimuli, standing in stark contrast with the standard visual neuroscience approach of presenting a gray 'blank' during an ISI. ISIs allow neurons to return to their baseline firing rate between visual stimuli to isolate responses with their corresponding stimulus; responses to stimuli that lack an ISI can therefore depend on the cell's response to the previous input. Under natural conditions, visual stimulation is a continuous input without ISIs. Our finding that responses to rapidly shifting full-field stimuli recapitulate gaze-shift responses supports the notion that neural dynamics depend on not only what stimuli are presented, but how they are presented. Accordingly, it is

important to understand the visual stimulus as it arrives at the retina under natural conditions⁴.

For most neurons the typical temporal response sequence around gaze shifts was replaced by suppression in the dark. This result is consistent with physiological saccadic suppression via corollary discharge, which has been reported in other species, including nonhuman primates and cats^{37–39}. It is possible that this suppressive signal is present in the light but is masked by the strong visually driven responses that are absent in the dark. It may also contribute to the negative component observed in gaze shifts in the light, amplifying the visually driven component. Alternatively, previous work in mice has also shown that responses to head rotation in V1 are luminance-dependent in a cell-type-specific manner¹⁷, and thus the sign of this signal may switch between the light and dark conditions. We also observed a small population of neurons (3.3%) that showed positive responses to gaze-shifting movements in both the light and dark, which may represent a direct motor corollary discharge signal. Similar results have been reported in head-fixed mice in the absence of visual input¹¹, though we observed a much smaller fraction of units with such responses.

Coarse-to-fine processing during active vision

There is substantial evidence from human psychophysics that supports coarse-to-fine processing, with lower SF components being recognized before higher SF components^{40–42}. At the neural level, single-unit recordings in head-fixed nonhuman primates and rodents have shown a progression of tuning from lower to higher SF following stimulus onset^{43–47}. Here, we demonstrate that this coarse-to-fine visual mechanism is engaged during active vision in both mice and marmosets, resulting in a temporal sequence of responses to visual input with gaze shifts.

There are multiple proposed reasons for why the visual system may utilize a coarse-to-fine order of processing²⁰. According to Marr's primal sketch theory, initial visual processing may be devoted to setting a basic layout of the visual scene, followed by later activity to establish greater dimensionality in the percept⁴⁸. Alternately, the order of responses may represent a matching to the typical input following a saccade and subsequent fixation, with lower SFs predominant immediately after a saccade due to the 1/f frequency distribution of natural scenes, followed by a whitening transformation of the distribution due to small movements during the fixations⁴⁹. Additionally, recent work in rodents suggests that coarse-to-fine processing may support efficient coding of natural scenes⁴⁴, and modeling shows that a coarse-to-fine framework improves binocular disparity computation in stereovision⁵⁰.

The underlying mechanisms that generate coarse-to-fine processing in V1 are unknown. One possibility is the pooling in V1 of magnocellular and parvocellular type inputs from the lateral geniculate nucleus, which respond in both a temporally and spatially specific manner⁵¹. In primates, the magnocellular pathway conveys high TF/low SF information ('fast' and 'coarse'), while the parvocellular pathway shows low TF/high SF tuning ('slow' and 'fine')⁵¹. Although there isn't a clear delineation of magnocellular and parvocellular pathways in mice, a similar relationship between transient/sustained response types and corresponding SF tuning has been observed in mouse lateral geniculate nucleus⁵², and previous work supports a relationship between TF and SF tuning in mouse V1 (ref. 53). Alternatively, a rapid refinement of SF tuning could result from recurrent processing in cortex, for example, from active suppression of low SF responses⁴³. Genetic tools available in mice may allow experiments to test these potential circuit mechanisms.

Together, these results present a dynamic view of processing during active vision, where each gaze shift initiates a temporal sequence supporting coarse-to-fine processing. This processing mode is shared across rodents and primates and is dramatically different from the regime typically probed in head- and gaze-fixed vision. This emphasizes the importance of accounting for the actual dynamics of visual input

that occur during behavior under natural conditions, to understand the computations performed by the visual system.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41593-023-01481-7>.

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Methods

Mouse methods

Animals. All procedures were conducted in accordance with the guidelines of the National Institutes of Health and were approved by the University of Oregon Institutional Animal Care and Use Committee. The 3–8-month-old adult mice (*Mus musculus*, C57BL/6J, Jackson Laboratories and bred in-house) were kept on a 12-h light/dark cycle. In total, eight male and six female mice were used for this study. Mice were housed with sibling cagemates until electrophysiology implant when they were singly housed. Humidity was between 40% and 60% and temperature was $21 \pm 1^\circ\text{C}$. Data collection and analysis were not performed blind to the conditions of the experiments. No animals or data points were excluded from analysis. Data collection was not randomized as there was only a single experimental group.

Surgery and habituation. Mice were initially implanted with a titanium headplate over V1 to allow for head-fixation and attachment of head-mounted experimental hardware. After 3 days of recovery, widefield imaging⁵⁴ was performed to help target the electrophysiology implant to the approximate center of the left monocular V1. A miniature connector (Mill-Max 853-93-100-10-001000) was secured to the headplate to allow repeated, reversible attachment of a camera arm, eye/world cameras and IMU^{21,22}. To simulate the weight of the real electrophysiology drive for habituation, a ‘dummy’ electrophysiology drive was glued to the headplate. Animals were handled by the experimenter for several days before surgical procedures, and subsequently habituated (~45 min total) to the spherical treadmill and freely moving arena with hardware tethering attached for several days before experiments.

The electrophysiology implant was performed once animals moved comfortably in the arena. A craniotomy was performed over V1, and a linear silicon probe (64 or 128 channels, Diagnostic Biochips P64-3 or P128-6) mounted in a custom three-dimensionally printed drive (Yuta Senzai, UCSF) was lowered into the brain using a stereotax to an approximate tip depth of 750 μm from the pial surface. The surface of the craniotomy was coated in artificial dura (Dow DOWSIL 3-4680) and the drive was secured to the headplate using light-curable dental acrylic (Unifast LC). A second craniotomy was performed above the left frontal cortex, and a reference wire was inserted into the brain. The opening was coated with a small amount of sterile ophthalmic ointment before the wire was glued in place with cyanoacrylate. Animals recovered overnight and experiments began the following day.

Hardware and recording. The camera arm was oriented approximately 90 deg to the right of the nose and included an eye-facing camera (iSecurity1011000TVL NTSC, 30 frames per s (fps) interlaced), an infrared light-emitting diode to illuminate the eye (Chanzon, 3-mm diameter, 940-nm wavelength), a wide-angle camera oriented toward the mouse’s point of view (BETA FPV C01, 30 fps interlaced) and an inertial measurement unit acquiring three-axis gyroscope and accelerometer signals (Rosco Technologies; acquired at 30 kHz, downsampled to 300 Hz and interpolated to camera data). Fine-gauge wire (Cooner, 36 AWG, no. CZ1174CLR) connected the IMU to its acquisition box, and each of the cameras to a USB video capture device (Pinnacle Dazzle or StarTech USB3HDCAP). A top-down camera (FLIR Blackfly USB3, 60 fps) recorded the mouse in the arena.

The electrophysiology headstage (built into the silicon probe package) was connected to an Open Ephys acquisition system via an ultra-thin cable (Intan no. C3216). Electrophysiology data were acquired at 30 kHz and bandpass filtered between 0.01 Hz and 7.5 kHz. We first used the Open Ephys GUI (<https://github.com/open-ephys/plugin-GUI>) to assess the quality of the electrophysiology data, then recordings were performed in Bonsai⁵⁵ using custom workflows (<https://github.com/nielllab/FreeMovingEphys>). System timestamps were collected for all hardware devices and later used to align data streams through interpolation.

During experiments, animals were first head-fixed on a spherical treadmill to permit measurement of visual tuning properties using traditional methods, then were transferred to an arena where they could freely explore. Recording duration was approximately 40 min head-fixed, and 1 h freely moving. For head-fixed experiments, a 70-cm monitor (BenQ GW2780) was placed approximately 27.5 cm from the mouse’s right eye, and visual stimuli were presented using Psychtoolbox-3 (ref. 56). Head-fixed stimuli were also recorded using the head-mounted world camera. First, we presented 15 min of a band-limited Gaussian noise stimulus⁵⁷ (SF spectrum 0.05 cpd to 0.12 cpd, flat TF spectrum with a low-pass cutoff at 4 Hz) to confirm V1 targeting based on spike-triggered average receptive fields. We presented a contrast-reversing square-wave checkerboard stimulus for 2 min with an SF of 0.04 cpd and TF of 0.5 Hz. Flashed sparse noise consisted of full- and minimum-luminance circular spots on a gray background played for 5 min. Spots were 2, 4, 8, 16 and 32 deg in diameter and presented so that each size made up an equal fraction of the area on the screen, totaling 15% on average. Each stimulus frame was presented for 250 ms and immediately followed the previous frame with no ISI⁵². Drifting sinusoidal gratings were presented at eight evenly spaced directions of motion for three SFs (0.02, 0.08, 0.32 cpd) and two TFs (2, 8 cps) for 15 min, with a 1-s stimulus duration and 0.5-s gray ISI, and stimulus conditions randomly interleaved (total of 12 presentations per unique stimulus).

For experiments comparing stimuli flashed with and without an ISI, responses were recorded for two stimulus sets: a sparse noise stimulus presented for 500 ms followed by a gray screen ISI lasting 500 ms, shown for 20 min (~1,200 stimulus frames). Next, the same stimulus flashed every 500 ms with no ISI interleaved between frames, shown for 10 min total (~1,200 stimulus frames).

The arena was approximately 48-cm long by 37-cm wide by 30-cm high. A 61-cm monitor (BenQ GW2480) covered one wall of the arena, while the other three walls were clear acrylic covering custom wallpaper including black and white high- and low-SF gratings and white noise. During recordings, the arena’s stimulus monitor played a sparse moving noise stimulus similar to the flashed sparse noise stimulus used during head-fixed recordings⁵². For moving sparse noise, full- and minimum-luminance spots were 4, 8 and 16 deg in diameter. Each spot was assigned to move in one of eight evenly spaced directions and at one of five speeds (10, 20, 40, 80, 160 deg s⁻¹). Spots appeared on the appropriate edge of the screen and moved across until they disappeared on the far edge. The floor was a gray silicone mat (Gartful) and was densely covered with black and white Lego bricks to provide three-dimensional visual contrast. Small pieces of tortilla chips (Juanita’s) were lightly scattered around the arena to encourage foraging during the recording; however, animals were not water or food restricted.

To eliminate all possible light within the arena during dark recordings, the entire behavioral enclosure was sealed in light-blocking material (Thorlabs BK5), all potential light sources within the enclosure were removed and all external light sources were turned off. Animals were first recorded in the dark (~20 min), then the arena lights and wall stimulus monitor were turned on (~20 min). As a result of the dark conditions, the pupil dilated beyond the margins of the eyelids, which made eye tracking infeasible. Thus, before the experiment, one drop of 2% pilocarpine HCl ophthalmic solution was applied to the animal’s right eye to constrict the pupil to a size similar to that seen in the light. Once the pupil was restricted enough for tracking in the dark (~3 min) the animal was moved into the dark arena for recording until the effects of the pilocarpine wore off (~20 min), at which time the light recording began.

Data preprocessing. The raw electrophysiology data from head-fixed and freely moving recordings acquired in the same session were concatenated into a single file for spike sorting, to allow us to confidently track single units across the entire experiment. Common-mode noise was

removed by subtracting the median across all channels at each time-point. Spike sorting was performed using Kilosort 2.5 (<https://github.com/MouseLand/Kilosort>), and isolated single units were then selected using Phy (Phy 2.0 beta 5; <https://github.com/cortex-lab/phy>) based on parameters including contamination (<10%), firing rate (mean > 0.5 Hz across entire recording), waveform shape and autocorrelogram. After data were spike sorted in Kilosort and curated in Phy, the resulting spike data were then split back out into individual recordings for analysis.

To group units into narrow- and broad-spiking putative cell types, we first normalized the mean spike waveform of each unit using the equation below, where W is the waveform and W_b is the mean of the first 200 μ s. We performed k -means clustering on the normalized waveforms ($k = 2$). The resulting clusters closely resembled the segregation of narrow- and broad-spiking units in previous studies based on explicit waveform criteria⁵⁷.

$$W_{\text{norm}} = (W - W_b) / (-\min(W - W_b))$$

Laminar depth was calculated from the multi-unit local field potential (LFP) of head-fixed neural data. The mean-centered LFP of each channel for a given shank of the linear silicon probe was bandpass filtered (1 to 300 Hz) and the power for the channel was calculated. The position of the maximum LFP power along the ordered sites of each shank was considered the center of layer 5 of cortex⁵⁸.

$$\text{Power}_{\text{ch}} = \text{sqrt}(\text{mean}(\text{LFP}_{\text{ch}}^2))$$

We extracted pupil position from the eye camera data²¹. Briefly, eye camera data were first deinterlaced to achieve 60-fps video, then eight points around the pupil were tracked with DeepLabCut⁵⁹. We then fit an ellipse to these eight points and computed pupil position in terms of angular rotation. The head-mounted world camera was deinterlaced to achieve 60-fps video and distortions from the camera lens were corrected using OpenCV. The mouse's position in the arena was tracked with DeepLabCut using top-down camera frames. We defined running as periods when the animal's neck point had a velocity >2 cm s⁻¹ and a stationary state as periods when the neck point had a velocity <2 cm s⁻¹.

Horizontal head rotation velocity was extracted from the IMU, converted to deg s⁻¹ and interpolated to eyecam frame timestamps. We defined leftward and rightward directions as the direction of head movement from the animal's perspective. For example, a movement of the right eye in the nasal direction would be a leftward movement. To select eye/head movement onsets, we first determined if the head velocity was >60 deg s⁻¹ in the leftward or rightward direction. If a sufficiently large head movement was made, we then separated it into either gaze-shifting or compensatory movements using gaze velocity, where gaze is defined as the sum of horizontal eye and head velocities²¹. If there was a high gaze velocity (>240 deg s⁻¹) concurrent with the head movement, the movement was considered gaze-shifting. Those that resulted in low gaze velocity (<120 deg s⁻¹) were considered compensatory. Eye/head movements that resulted in intermediate gaze velocities (>120 deg s⁻¹ and <240 deg s⁻¹) were excluded from our analysis to avoid contamination between the two categories of movements (Extended Data Fig. 1e–i). For eye/head movements spanning multiple eye camera frames, only the first timepoint, representing the onset of the movement, was used to calculate the onset of the PETH. Compensatory movements which occurred 250 ms before or after a gaze-shifting movement were excluded to avoid contamination by gaze shifts. For head-fixed recordings, only high eye velocity (>240 deg s⁻¹) was used to identify saccades, since the head was immobilized.

To maintain internal consistency with eye cameras and obviate the need for a separate video stimulus synchronization signal, we determined stimulus onsets in head-fixed recordings directly from the head-mounted cameras. For the reversing checkerboard stimulus

that updated every 500 ms, we identified frame transitions from the head-mounted camera video based on k -means clustering ($k = 2$) of the video into the two separate contrasts, and selected transitions between the clusters. For the flashed sparse noise stimulus that updated every 250 ms, we determined the timestamps of stimulus onset based on the root mean square pixel-wise change in the image, which showed clear peaks at stimulus transition. For drifting sinusoidal gratings, we determined direction of motion by computing optic flow from the worldcam video, SF based on the mean gradient magnitude from paired Sobel operators and TF from the mean Fourier transform of each pixel over time.

Analysis of neural responses. All analyses were performed in Python (v.3.8, python.org). Neural responses were calculated as a PETH from electrophysiology spike times using kernel density estimation with a gaussian kernel with bandwidth (standard deviation) of 10 ms, sampled at 1-ms intervals. The PETH included neural activity from -1,000 ms to 1,000 ms around the event onset for all stimuli/events (unless noted otherwise). A modulation index was calculated for PETHs as the peak of the firing rate (R) from the event onset at 0 ms until 250 ms after the event, minus the baseline before the event (R_b), which was calculated as the mean of R from -1,000 ms to 200 ms around the event onset (unless noted otherwise).

$$\text{mod} = \max(\text{abs}(R - R_b))$$

For the drifting gratings stimulus, the PETH was calculated similarly to other stimuli, but from -1,500 ms before until 1,500 ms after stimulus onset to capture the full stimulus and ISI interval.

The modulation index for gaze-shift responses in the leftward and rightward directions were compared to determine a preferred direction. A direction selectivity index (DSI) was calculated as the difference between the preferred and nonpreferred maximum modulations divided by their sum.

$$\text{DSI} = (\text{mod}_{\text{pref}} - \text{mod}_{\text{nonpref}}) / (\text{mod}_{\text{pref}} + \text{mod}_{\text{nonpref}})$$

To normalize PETHs, we subtracted the baseline R_b from the PETH (R) to give an evoked firing rate, and then divided by the maximum of R during a response window (R_{rw}) within a response range of -250 ms before to 250 ms after the event. The denominator includes responses up to 250 ms before the event to ensure that spurious increases in firing rate that are not time-locked to an event were not amplified by the normalization (this was extremely rare and only occurred in unresponsive cells).

$$R_{\text{norm}} = (R - R_b) / (\max(R_{\text{rw}}))$$

When normalizing the PETH of compensatory eye/head movements, we used that neuron's R_{rw} from gaze shifts in the preferred left/right direction so that all eye/head responses were normalized relative to the neuron's best response. Because the head-fixed sparse noise stimulus presentation was of short duration (250 ms), we used the spike rate at the time of the stimulus onset, 0 ms, as the baseline firing rate (R_b) for normalization to avoid including responses to previously presented flashed stimuli.

Neurons were considered responsive to a stimulus or eye/head movement onset if they changed their firing rate by at least 10% and 1 spike per second (sp s⁻¹) in the 250 ms following the onset of the event. For the gratings stimulus, suppressed-by-contrast cells were not considered responsive, and removed by requiring that gratings PETHs have a peak firing rate >0.5 normalized sp s⁻¹ during the 1 s of stimulus presentation.

The latency of a neuron's peak response was calculated as the timepoint of the maximum firing rate of its normalized PETH in the

period between 25 ms to 250 ms after the onset of the event. For temporal sequence plots, normalized PETHs were sorted by the latency of their peak for the preferred direction of gaze-shifting eye/head movements. Peak latency was not calculated for units unresponsive to gaze shifts. Peak latency sorting was cross-validated by randomly assigning gaze-shift events into a train set and a test set, calculating a PETH for each half of the data, and sorting the test set by the peak times of the training set (Extended Data Fig. 3).

SF and TF preferences were based on the evoked firing rate for each stimulus condition, computed as the mean rate from 25 to 1,000 ms following stimulus onset, minus the mean baseline rate in the 500 ms before stimulus onset. The mean SF and TF tuning curves for all cells within gaze-shift clusters were calculated using each cell's mean evoked firing rate for each SF/TF normalized by the maximum response to its best SF/TF.

Each cell's weighted TF response, W_{TF} , was determined using the mean evoked responses, R , for each of the two presented TFs and at the cell's optimal SF and preferred orientation.

$$W_{TF} = (2 \times R_{low} + 8 \times R_{high}) / (R_{low} + R_{high})$$

Likewise, weighted SF response was calculated at the cell's weighted TF response and preferred orientation using the mean evoked responses for each of the three presented SFs.

$$W_{SF} = (0.02 \times R_{low} + 0.08 \times R_{mid} + 0.32 \times R_{high}) / (R_{low} + R_{mid} + R_{high})$$

As an alternate metric for TF and SF bias in each gaze-shift cluster, we calculated the fraction of cells with a >2:1 preference for one of the presented SFs over the other two SFs (Extended Data Fig. 5c) and for one TF over the other (Extended Data Fig. 5g).

Response clusters. To cluster units with similar gaze-shift responses, we first performed PCA on the normalized gaze-shift responses in each unit's preferred direction in the period from -50 to 300 ms relative to the gaze-shift onset. Units that did not meet the above spike rate criteria had their input for PCA replaced with an array of zeros to exclude their nonresponsive PETHs from clustering, while seeding the formation of an unresponsive group of units for clustering. We used the first four principal components, which cumulatively explained 95% of the variance, to perform k -means clustering ($k = 5$), resulting in four clusters of responsive units and one cluster of unresponsive units. To maintain consistency, the same PC weights and k -means model were applied to cluster the smaller number of units that were acquired subsequently in light/dark experiments.

To identify units that respond positively in dark conditions, we selected units that were positively modulated in the dark by the onset of a gaze shift by $>1 \text{ sp s}^{-1}$ and 10% of their normalized spike rate in the 100 ms following the gaze-shift onset.

For presentation of sparse noise with and without an ISI (Extended Data Fig. 6a,b), neurons were grouped into response types by applying the PCA and k -means models fit with separate gaze-shift PETH data to the PETHs of the continuously presented flashed sparse noise. Cells were grouped for the ISI condition using the results of their clustering of the continuous stimulus. To avoid the inclusion of suppressed-by-contrast cells, we removed ISI-condition responses that failed to increase their firing rate by 0.1 normalized sp s^{-1} in the first 100 ms after stimulus onset.

Linear-nonlinear model. To model the temporal dynamics of responses to a time-varying stimulus with abrupt transitions, such as gaze shifts or flashed stimuli, we implemented a rate-based linear-nonlinear model⁶⁰. The stimulus was described by a set of discrete levels $s(t)$, and a neuron's temporal kernel as $k(\tau)$, each on a discrete timebase with $dt = 25 \text{ ms}$. The neuronal response at each time t was computed

as the convolution of the temporal kernel with the stimulus, followed by an exponential nonlinearity.

$$r(t) = \exp \left(\sum_{\tau=1}^n s(t-\tau)k(\tau) \right)$$

For each 500-ms interval the stimulus was a randomly selected integer in the range 1–8, either with or without a subsequent 500-ms ISI with a value of 0. For the temporal kernel, we use (a) the positive phase of a sine function, for a unimodal positive kernel; (b) the full cycle of a sine function, for a biphasic kernel; or (c) the full cycle of a sine function with the negative phase multiplied by 0.25, for an intermediate between a and b. Each kernel had a total duration of 250 ms.

Marmoset methods

Animals and surgical procedures. All surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee at the University of Rochester in accordance with the US National Institutes of Health guidelines. Two adult male common marmosets (*Calithrix jacchus*) aged 2 and 7 yr were used in this study. Housing temperature was 27 °C and humidity 50%. Data collection and analysis were not performed blind to the conditions of the experiments. No animals or data points were excluded from analysis. Data collection was not randomized as there was only a single experimental group. Marmosets underwent an initial surgery to implant a headpost to stabilize their head during behavioral sessions^{61,62}. A second surgery was performed to implant a recording chamber with a $3 \times 3\text{-mm}^2$ craniotomy over foveal and parafoveal V1. The exposed dura was sealed under a thin layer ($<1 \text{ mm}$) of a silicone elastomer⁶³.

Hardware and recordings. Electrophysiological recordings were performed using 2×32 -channel silicon electrode arrays (<http://www.neuronexus.com>). Probes included two sharpened tip shanks of 50- μm width spaced 200 μm apart, each containing 32 channels separated by 35 μm . In one animal we used a semichronic microdrive (EDDS Micro-drive system, <https://micropobes.com>) to place electrodes in cortex for 1–2 weeks over which we made 3–6 recordings. In the second animal we used a custom microdrive (<https://marmolab.bcs.rochester.edu/resources/>) to place and remove electrodes daily. Arrays were lowered slowly through silastic into cortex using a thumb screw.

Data were amplified and digitized at 30 kHz with Intan headstages using the Open Ephys GUI (<https://github.com/open-ephys/plugin-GUI>). The wideband signal was high-pass filtered by the headstage at 0.1 Hz, preprocessed by common-average referencing across all channels and then high-pass filtered at 300 Hz. The resulting traces were spike sorted using Kilosort2. Outputs from the spike sorting algorithms were manually labeled using the Phy GUI (<https://github.com/kwikteam/phy>). Any units that were either physiologically implausible based on the lack of a waveform with a trough followed by a peak or with an interspike interval distribution with more than 1% of the spikes under 1 ms were excluded from analyses.

Eye-tracking and saccade detection. Gaze position was monitored using infrared eye-tracking methods⁶⁴. Briefly, the first and fourth Purkinje images (P1 and P4) were visualized using a collimated infrared light source and tracked at 593 fps to estimate the two-dimensional eye angle. The eye tracker was manually calibrated to adjust the offset and gain (horizontal and vertical) by showing marmoset monkeys small windowed face images at different screen positions to obtain their fixation^{61,62}. Saccadic eye movements were identified automatically using a combination of velocity and acceleration thresholds⁶⁵.

Head-fixed visual stimuli and free-viewing tasks. Visual stimuli were presented on a Propixx Projector (Vpixx) with a linear gamma table using Psychtoolbox-3 (ref. 56) and Matlab (MathWorks). Stimulus and

physiology clocks were aligned using a Datapixx (Vpixx)⁶⁶. Marmov5 stimulus code is available (<https://github.com/jcbyts/Marmov5>).

Marmosets performed free-viewing and visual-foraging tasks. For natural image stimuli, the marmosets were allowed to free-view a screen containing a full-field (± 15 visual degrees) grayscale natural image for 10 s and then received a single drop of juice (5–10 μ l of marshmallow water). On the foraging trials the marmosets viewed a random sequence of flashed full-field gratings (60- or 120-Hz frame update of varying orientation and SF gratings) as they searched for a small Gabor target⁶⁴. A drop of juice reward was given for fixating within 2 deg of the small target for more than 100 ms after which time it was repositioned to another random location within 5 visual degrees of the screen center. During foraging the full-field flashed gratings were presented at 25% contrast and were drawn from a polar grid of 8 equally spaced orientations and a grid of 4 log-spaced range of SFs from either 1 to 8 cpd or 2 to 16 cpd (8×4 stimulus space). On each video frame either a single grating was presented or with 50% chance a blank gray background was shown.

Estimation of orientation and SF tuning. To measure orientation and SF preferences of individual neurons we analyzed the response to flashed full-field sine-wave gratings described above for the foraging task. A linear estimate of the receptive field can be recovered from this stimulus using subspace reverse correlation by correlating the firing rate during each video frame with the stimulus history²⁹. We first computed a temporal kernel of the response to gratings as compared with blanks. The average temporal response at latencies from –50 to 200 ms (–3 to 12 video frames at 60 Hz) was computed by time-locking on video frames including gratings and averaging. It was compared against the baseline firing rate, which was estimated by instead time-locking to blank video frames. If a peak was identified in the temporal response to the grating stimulus between 20 and 80 ms that was at least 8 s.d. above the baseline, then the neuron was included in analyses for tuning. We marked a temporal window that included the frame of the peak response along with the adjacent frames next to the peak frame if they were significant by at least five standard deviations above the baseline.

To estimate orientation and SF tuning, the mean response was computed from the identified temporal window for each grating stimulus. The joint tuning for orientation and SF was defined by the mean responses in the 8×4 grid of stimuli sampled from which we averaged across orientation or SF to get the marginal tuning curves. To determine if neurons exhibited a selective response, we computed a tuning index by taking the standard deviation of the values in the tuning curve divided by the mean. Neurons with weak tuning, having an index less than 0.2 in both dimensions, were not included for subsequent analyses that aimed to correlate tuning preference with saccade modulation. The weighted preference was computed using a weighted mean from the tuning curve instead of taking the peak at a discrete value, similar to the analysis for mouse data described above. To emphasize the peak in the weighted mean, since a wider range of SFs was presented than for mouse data, we used a weighting function computed by subtracting the mean of the tuning curve with the resulting negative values floored at zero and then squaring the remaining nonzero values.

Neural responses to eye movements. Neural responses around eye movements were computed from –200 ms before to 400 ms after the saccadic onset. The PETH was computed using kernel density estimation with a Gaussian kernel with half-bandwidth of 10 ms, sampled at 1-ms intervals. To normalize PETHs (R), we computed the baseline firing rate (R_b) from –100 to 0 ms before saccade onset and normalized the PETH as

$$R_{\text{norm}} = (R - R_b) / \max(R).$$

The maximum of R was computed from 0 to 400 ms. The PCA was computed on the normalized PETH from –200 to 400 ms. The six PCs explained 95% of the variance and were k -means clustered ($k = 4$) to group units into saccade response types.

Statistical analysis

Statistical tests across cell types were performed using ANOVA with correction for multiple comparisons or a linear mixed effects model. Correlations between latencies and response properties were tested using Spearman's correlation coefficient. Bar plots show mean and s.e.m. unless otherwise noted. No statistical methods were used to predetermine sample sizes, but our sample sizes are similar to those reported in previous publications^{19,67}. Data distribution was assumed to be normal, but this was not formally tested.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Data are available on Dryad: <https://doi.org/10.5061/dryad.kd51c5bck>. These are the minimum data required to reproduce the figures in this publication using the 'analysis and figure generation' code linked in the 'Code availability' section below.

Code availability

All original code is publicly available. Mouse data preprocessing: <https://github.com/nielllab/FreelyMovingEphys>; analysis and figure generation: <https://github.com/nielllab/freely-moving-saccades>; marmoset stimulus analysis: <https://github.com/jcbyts/Marmov5>.

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Author contributions

P.R.L.P. and C.M.N. conceived the project. P.R.L.P. and E.S.P.L. led mouse experiments. D.M.M. and C.M.N. led data analysis. N.M.C. and S.L.S. contributed to mouse experiments. E.T.T.A. contributed to data analysis. M.C.S. generated the audio track from mouse neural activity. J.L.Y. and J.F.M. performed marmoset experiments. J.L.Y., J.F.M. and

D.M.M. performed marmoset data analysis. All authors contributed to writing and editing the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

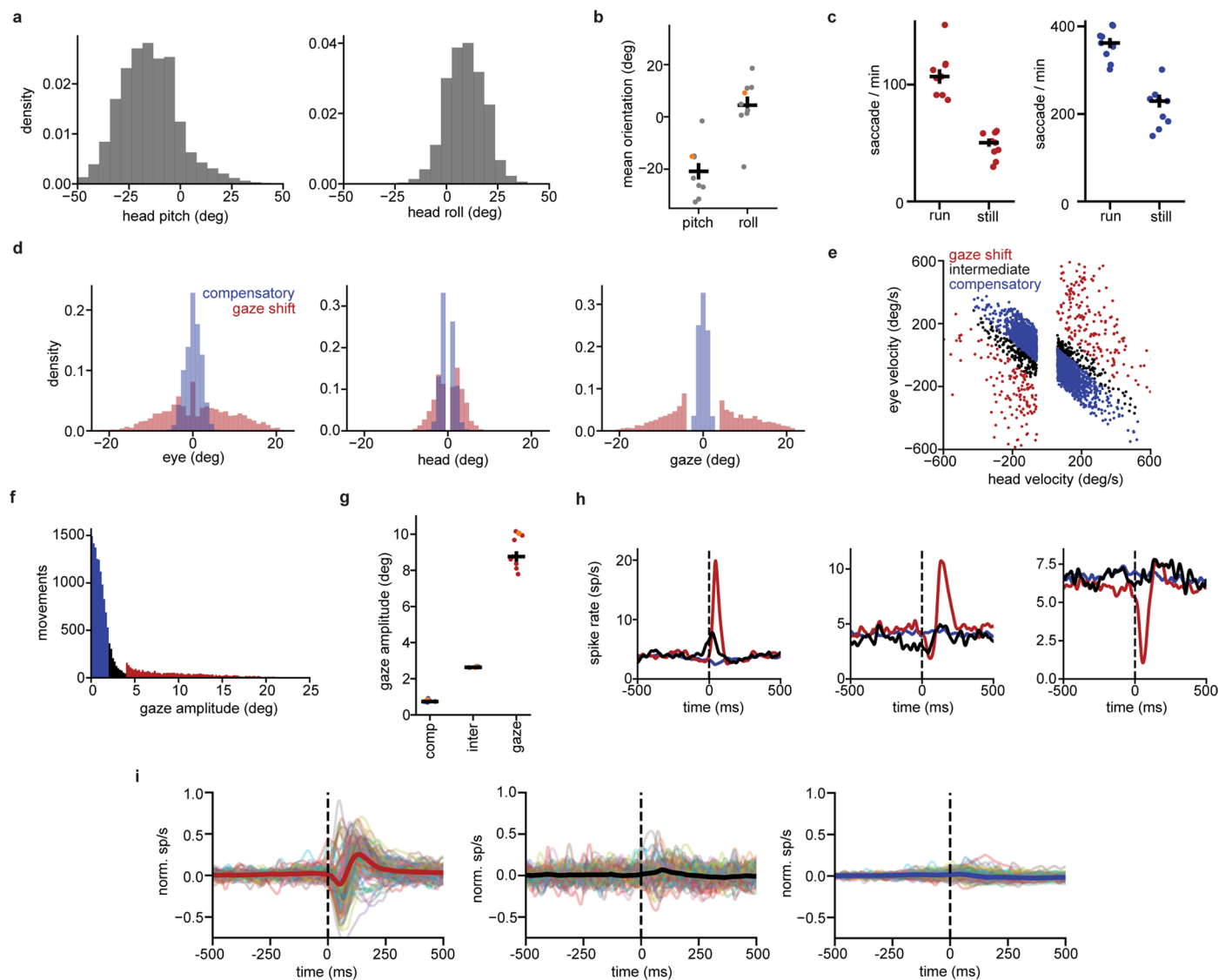
Extended data is available for this paper at <https://doi.org/10.1038/s41593-023-01481-7>.

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41593-023-01481-7>.

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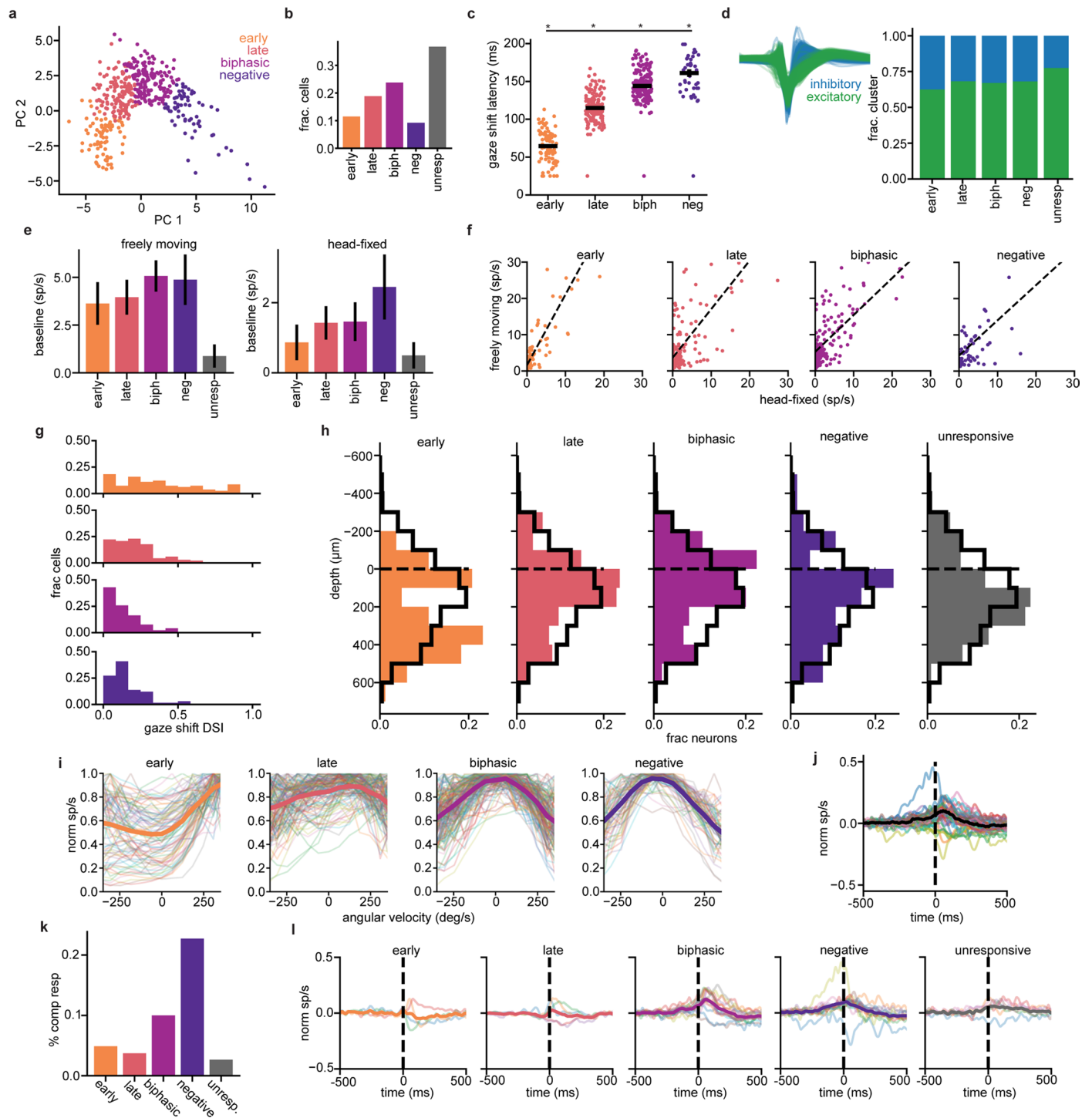
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Extended Data Fig. 1 | Characterization of free movement. **a.** Mean head pitch and roll during free motion for one example recording. Pitch mean = -15.1 ± 0.02 deg; Roll mean = 9.2 ± 0.01 deg. **b.** Mean head pitch and roll, indicating the center point during free movement. Each point is a mouse ($n = 9$ mice). Black bars indicate mean and standard error (pitch: -20.8 ± 3.1 deg; roll: 4.5 ± 3.3 deg). Recording from **a** shown in orange. **c.** Rate of gaze-shifting (left; run median = 106 ± 6 saccades/min; still median = 50 ± 3 saccades/min) and compensatory (right; run median = 362 ± 11 saccades/min; still median = 229 ± 15 saccades/min) movements during periods of locomotion greater than 2 cm/s measured from the top camera ('run') compared to periods of slower locomotion, fine motion, and/or stationary periods less than 2 cm/s ('still'). Each point is a mouse ($n = 9$ mice). Black bars indicate median and standard error. **d.** Amplitude of position change for eye (left), head (middle) and gaze (right; defined as eye

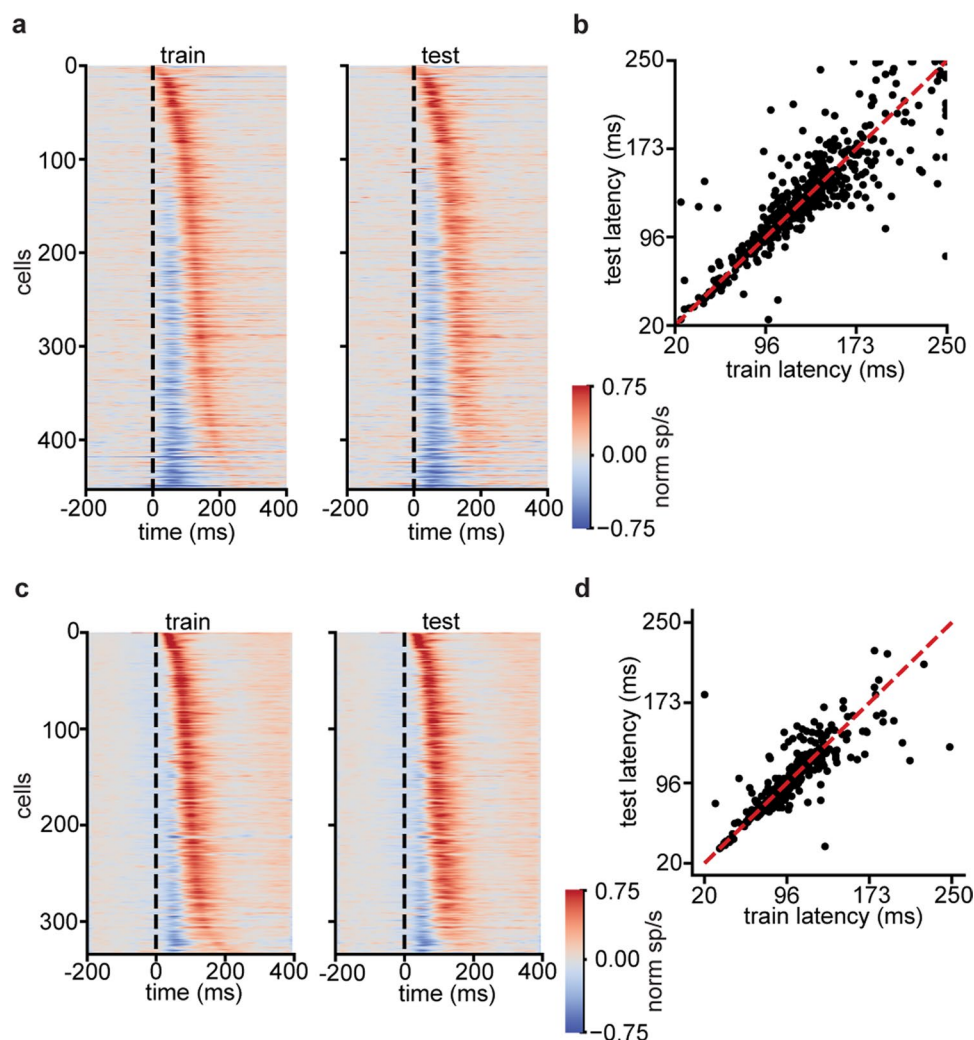
+ head) during gaze-shifting and compensatory eye/head movements at the onset of the movement for the example recording used in **a**. **e.** Scatter plot of eye and head velocities subsampled (25x) from the example recording used in **a**, showing compensatory, gaze-shifting, and intermediate movements, the latter of which are excluded from the analysis in the main text. **f.** Amplitude of gaze changes at onset of movement for the example recording used in **a**. **g.** Median $\pm$ SEM amplitude of gaze change for all recordings. Each recording is a point ($n = 9$ mice). Recording in **f** is shown in orange. Compensatory: 0.73 ± 0.02 deg; intermediate: 2.63 ± 0.01 deg; gaze-shifting: 8.76 ± 0.29 deg. **h.** PETHs for example cells from Fig. 1g including PETH for responses to intermediate saccades in black. **i.** Normalized PETHs of gaze-shifting (left), intermediate (middle), and compensatory (right) eye/head movements for 100 example units with a baseline firing rate >2 Hz, with median of all cells ($n = 716$) overlaid.



Extended Data Fig. 2 | See next page for caption.

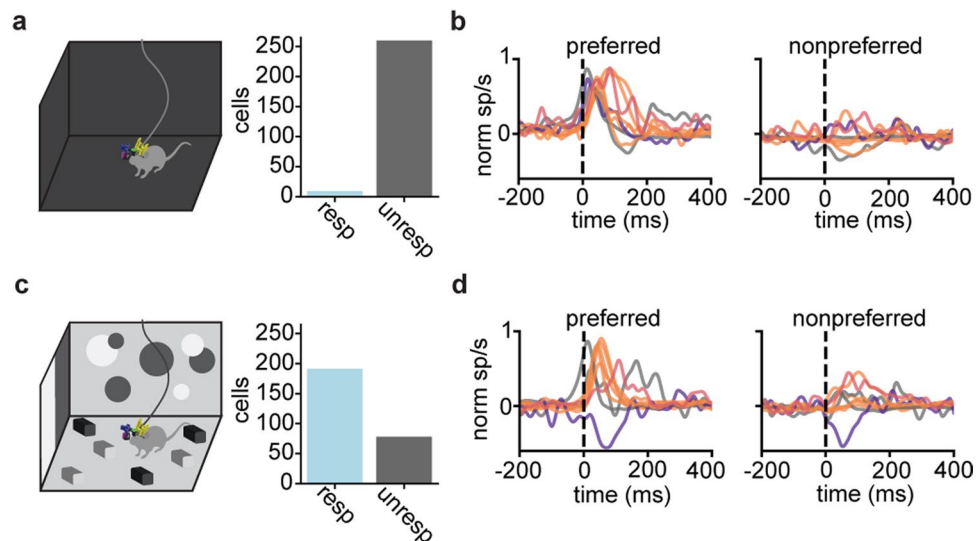
Extended Data Fig. 2 | Additional characterization of gaze shift response types. **a.** PCA of gaze shift PETHs. Only the two PCs with the highest explained variance are shown. Cells in the scatter plot are colored by the cluster they were assigned by k-means clustering of PCs. **b.** Fraction of units in each gaze shift response cluster. **c.** Latency of peak responses were significantly different for all comparisons between clusters ($p < 0.05$ with no effect of experimental session $p = 0.220$, linear mixed effects model; early vs. late $p = 7.10 \times 10^{-42}$, early vs. biphasic $p = 3.86 \times 10^{-116}$, early vs. negative $p = 3.83 \times 10^{-162}$, late vs. biphasic $p = 2.32 \times 10^{-25}$, late vs. negative $p = 2.30 \times 10^{-65}$, biphasic vs. negative $p = 6.67 \times 10^{-22}$). **d.** Fraction of putative cell types in each gaze shift response cluster. Excitatory and inhibitory groups were identified by k-means clustering on spike waveforms (waveforms shown above). **e.** Median $\pm$ SEM baseline firing rate of units during freely moving (left) and head-fixed (right) recordings ($n = 9$ mice, $n = 716$ cells). Freely moving baseline was calculated as the pre-saccadic period before gaze shifts. Head-fixed baseline was calculated as the firing rate during presentation of gray screen during head-fixation. **f.** Scatter plot of head-fixed and freely moving baseline

firing rates. Each point is a cell. Linear regression shown as dashed black line. (early: $r = 0.87$, $p = 1.01 \times 10^{-26}$, $m = 1.94$; late: $r = 0.70$, $p = 7.06 \times 10^{-21}$, $m = 1.33$; biphasic: $r = 0.70$, $p = 3.01 \times 10^{-26}$, $m = 1.01$; negative: $r = 0.69$, $p = 1.45 \times 10^{-10}$, $m = 0.97$). **g.** Gaze shift left/right direction selectivity index by cluster. **h.** Laminar depth of all cells determined using the local field potential from multi-unit activity power along each shank of the probe. Black outline shows the distribution of depths for all cells. Dashed line ($0 \mu\text{m}$) is the estimated depth of cortical layer 5, to which depths were aligned. **i.** Normalized horizontal angular velocity tuning for all cells, separated by response clusters. Positive values for angular velocity represent each unit's preferred horizontal direction of gaze shift. **j.** PETH for compensatory eye/head movements for cells responsive to compensatory movements ($n = 48/716$). Only the preferred direction is shown. Responsiveness defined as 10% modulation and modulation by at least 1 sp/s. **k.** Percent of each gaze shift response cluster that is responsive to compensatory movements (total=48/716, early=4/82, late=5/135, biphasic=17/170, negative=15/66, unresponsive=7/263) **l.** Same as **j** grouped by gaze shift response cluster.



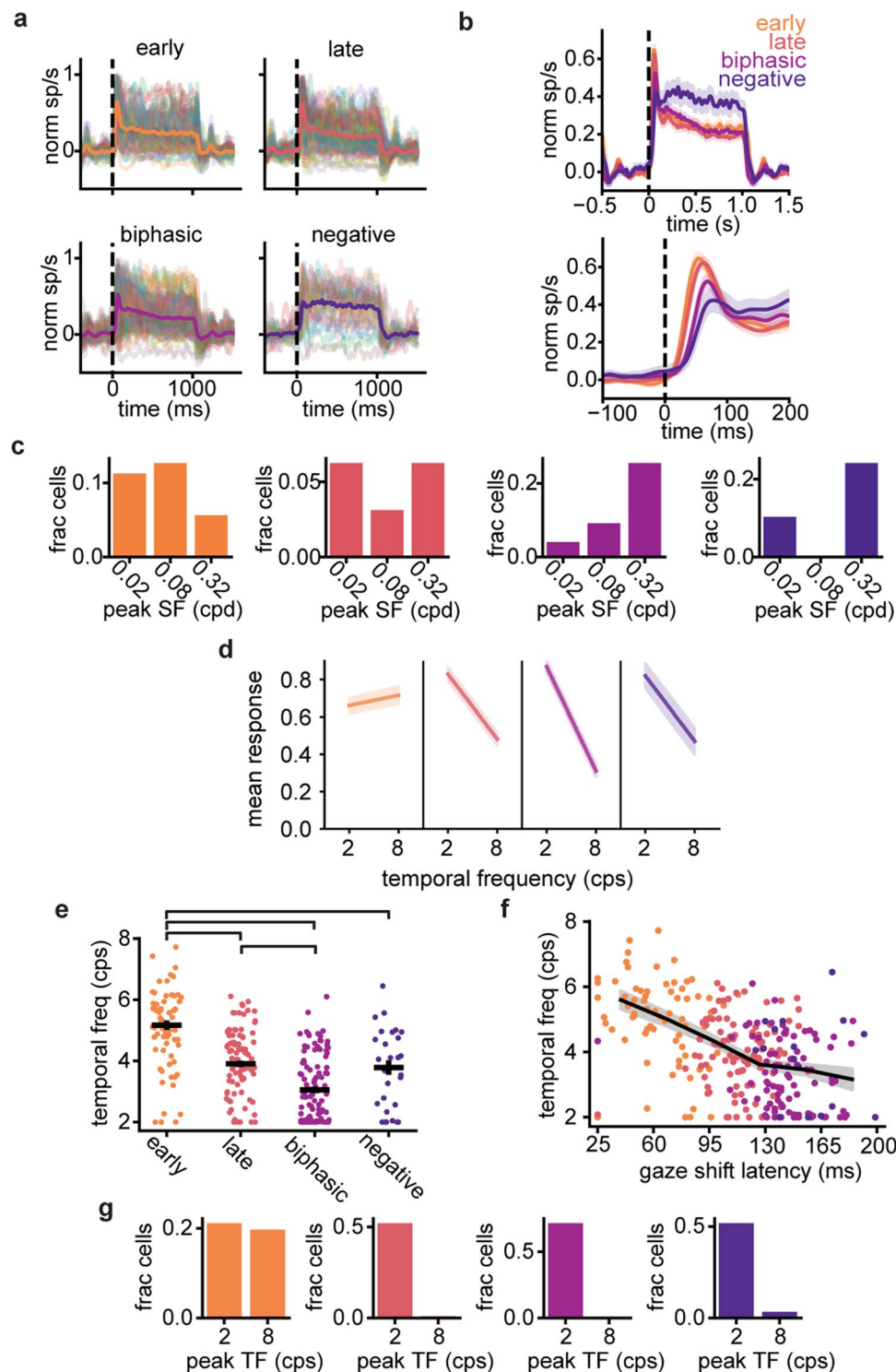
Extended Data Fig. 3 | Cross validation of response latencies. **a.** Cross-validation for mouse gaze shift PETHs of all responsive cells. Gaze shift times were randomly divided into two sets used to calculate PETHs in the train (left) and test (right) sets. The test set was sorted by the latency of the positive peak in

the train set. **b.** Latency of gaze shift response for train versus test sets ($r = 0.870$, $p = 2.51e-140$). **c.** Same as **a** for marmoset saccades. **d.** Same as **b** for marmoset saccades ($r = 0.875$, $p = 1.44e-106$).



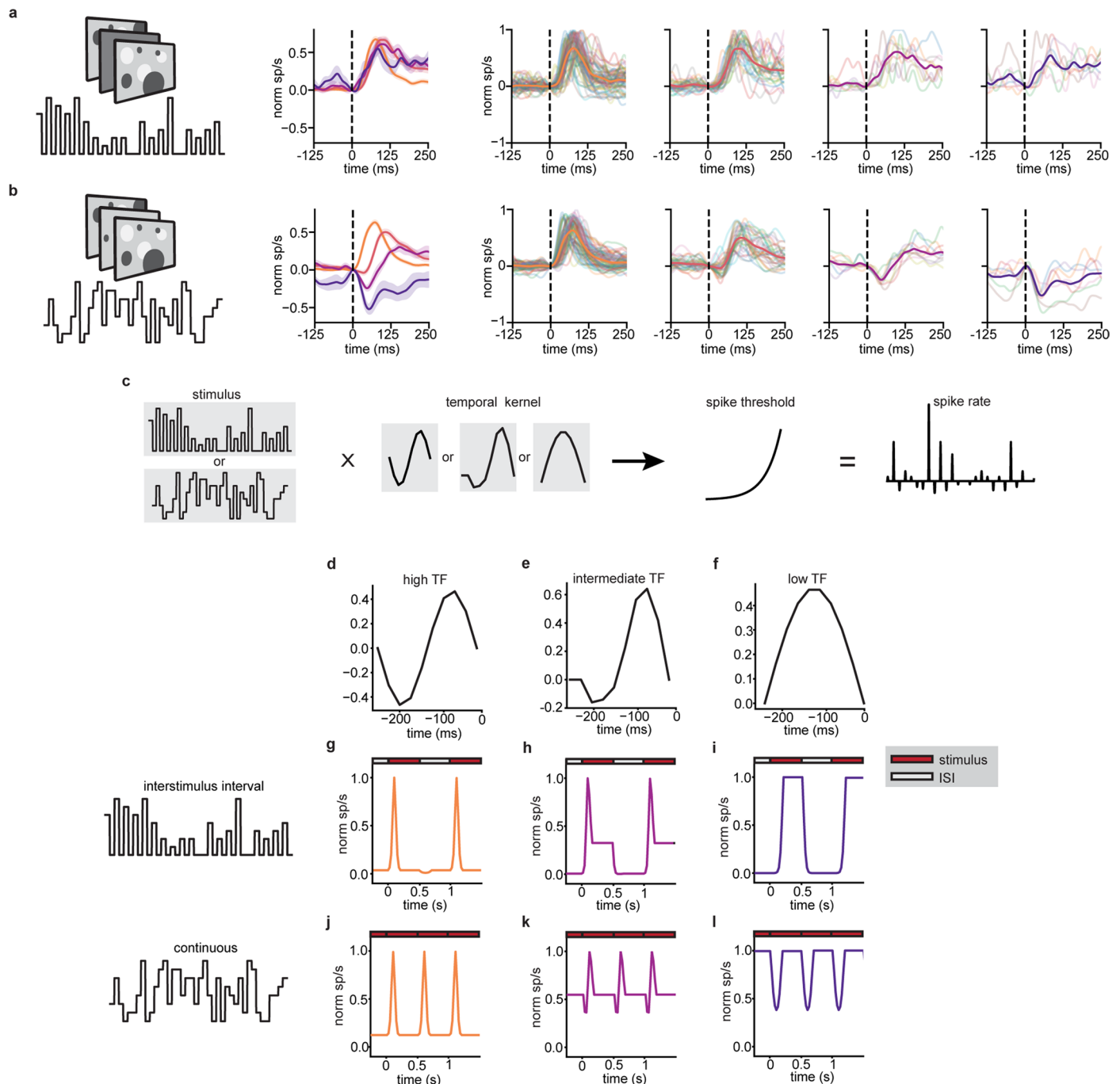
Extended Data Fig. 4 | Additional characterization of responses in the dark. **a.** Fraction of cells responsive in the dark condition (responsive=9/269, unresponsive=260/269). **b.** Dark condition PETHs for cells that responded to gaze shifts in freely moving dark conditions. Units are colored by clustering from

responses in light condition. $n = 9/269$ (early=5, late=1, biphasic=0, negative=1, unresponsive=2). **c.** Same as **a** for the light condition (responsive=191/269, unresponsive=78/269). **d.** Responses of units in **b** for the light condition.



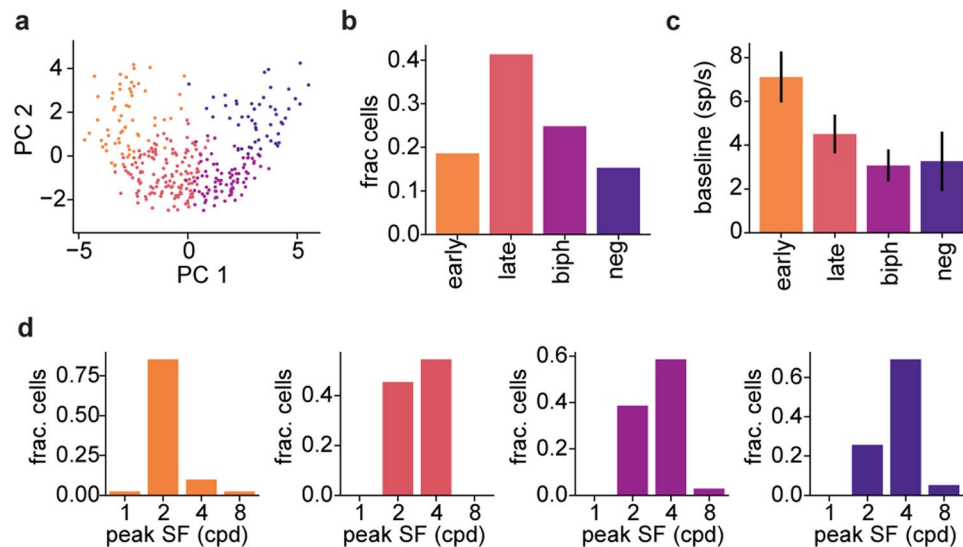
Extended Data Fig. 5 | Additional characterization of drifting gratings responses. **a.** Head-fixed drifting gratings PETHs for gaze shift response clusters with mean response overlaid. Stimulus is presented for 1 s with gray ISI between stimuli. $n = 9$ mice, $n = 384/716$ cells responsive to gratings (early=71, late=96, biphasic=98, negative=29, unresponsive=90). Cells below firing rate threshold are not shown. **b.** Mean $\pm$ SEM normalized gratings PETHs clustered by gaze shift response for full stimulus presentation (top) and highlighting stimulus onset (bottom). **c.** Fraction of cells in each cluster with a $\geq 2:1$ preference for the presented spatial frequencies compared to the sum of responses for the two other spatial frequencies. **d.** Mean $\pm$ SEM temporal frequency tuning curve by cluster (Multivariate two-way ANOVA, TF $\times$ cluster $F = 21.45$, $p = 3.45 \times 10^{-13}$).

e. Temporal frequency preference for gratings-responsive cells in each gaze shift response cluster, calculated as a weighted mean of responses ($n = 9$ mice, 384 cells). Median and standard error are shown for each cluster. Bars above indicate statistical significance at $p < 0.05$ (linear mixed effects model, $n = 9$ mice, $n = 384$ cells; early vs. late $p = 3.64 \times 10^{-7}$, early vs. biphasic $p = 2.24 \times 10^{-21}$, early vs. negative $p = 4.42 \times 10^{-9}$, late vs. biphasic $p = 2.32 \times 10^{-6}$, late vs. negative $p = 5.69 \times 10^{-2}$, biphasic vs. negative $p = 9.37 \times 10^{-2}$). **f.** Weighted temporal frequency preference versus gaze shift response latency, for all cells responsive to gratings. Running median $\pm$ SEM for all cells is overlaid. The color of each point indicates the cluster from gaze shift responses. ($r = -0.468$, $p = 2.12 \times 10^{-16}$). **g.** Same as **c** for temporal frequency.



Extended Data Fig. 6 | Temporal tuning of neurons can explain diverse responses to gaze shifts. **a.** Responses to flashed sparse noise stimulus presented with an inter-stimulus interval (ISI; $n = 3$ mice; early=71, late=33, biphasic=9, negative=7). Left: mean $\pm$ SEM for each cluster; right: individual neuron responses overlaid with mean. **b.** Same as **a** for a continuously flashed sparse noise stimulus. **c.** Schematic of modeling approach. A scalar stimulus, presented continuously or with an inter-stimulus interval (ISI), is passed

through a variable temporal kernel of either high, intermediate, or low temporal frequency (TF), and a non-linearity is used to generate a spiking output. **d.** High TF kernel. **e.** Intermediate TF kernel. **f.** Low TF kernel. **g.** Resulting response of model using high TF kernel to visual stimuli presented with an ISI. **h.** Same as **g** for intermediate TF kernel. **i.** Same as **g** for low TF kernel. **j.** Resulting response of model using high TF kernel to visual stimuli presented continuously. **k.** Same as **j** for intermediate TF kernel. **l.** Same as **j** for low TF kernel.



Extended Data Fig. 7 | Additional characterization of marmoset saccade response types. a. PCA of marmoset gaze shift PETHs for the 2 PCs with highest explained variance, colored by k-means clusters. **b.** Fraction of units in each

saccade response cluster. **c.** Median $\pm$ SEM baseline firing rate of units in each cluster ($n = 2$ marmosets, $n = 238$ cells). **d.** Fraction of cells with maximal response to each presented spatial frequency.

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- ☐ ☒ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
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Software and code

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Python 3.8 <https://www.python.org/>
Open Ephys plugin-GUI v0.6 <https://github.com/open-ephys/plugin-GUI>
Bonsai 2.6 <https://github.com/bonsai-rx/bonsai>
Custom Bonsai code <https://github.com/nielllab/FreelyMovingEphys>
MarmoV5 <https://github.com/jcbyts/MarmoV5>
Psychtoolbox-3 <http://psychtoolbox.org/>

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Python 3.8 <https://www.python.org/>
OpenCv 4.8 <https://pypi.org/project/opencv-python/>
Kilosort 2.5 <https://github.com/MouseLand/Kilosort>
Phy 2.0 beta 5 <https://github.com/cortex-lab/phy>
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Data have been deposited on Dryad: <https://doi.org/10.5061/dryad.kd51c5bck>

Human research participants

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Reporting on sex and gender

n/a

Population characteristics

n/a

Recruitment

n/a

Ethics oversight

n/a

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

The number of animals (sample size) was determined based on species-specific standards for replicability in neural recordings, with n=14 mice and n=2 marmosets. (Senzai & Scanziani, Science 2022; Liska et al., eLife 2023).

Data exclusions

No data was excluded from analysis.

Replication

Experiments were repeated in 14 mice and 2 marmosets with consistent results across animals.

Randomization

Subjects were not assigned to experimental groups, as all experiments were performed in wildtype animals.

Blinding

Subjects were not assigned to experimental groups, as all experiments were performed in wildtype animals.

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☒	☐ Eukaryotic cell lines
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☐	☒ Animals and other organisms
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Laboratory animals	Mus musculus, C57BL6J, 8 male and 6 female, three to eight months of age; Callithrix jacchus, 2 males, 2 and 7 years of age
Wild animals	This study did not involve wild animals.
Reporting on sex	Mouse: both sexes were used, but sex-based analyses were not performed due to insufficient statistical power. Mice were assigned randomly in equal numbers. Marmoset: two males were used based on limited availability of subjects.
Field-collected samples	The study did not involve samples collected from the field.
Ethics oversight	Mice: all procedures were conducted in accordance with the guidelines of the National Institutes of Health and were approved by the University of Oregon Institutional Animal Care and Use Committee. Marmosets: all surgical and experimental procedures were approved by the Institutional Animal Care and Use Committee at the University of Rochester in accordance with the US National Institutes of Health guidelines.

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